



ORIGINAL RESEARCH ARTICLE

Characterization of the Adhesive Layer through Finite Element Modeling in Double Strap Joints and Validation Through Digital Image Correlation

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The accurate measurement of strain on engineering structures is crucial for evaluating their performance and ensuring their structural integrity. Strain gauges are employed for this purpose, enabling to monitor mechanical deformations and stress distributions. The adhesive layer serves as a medium between the strain gauge and the material being tested. Changes in the adhesive layer can impact, and strain transfers between materials and strain gauges when subjected to different types of loading. The present study investigates the adhesive layer's role in strain gauge mounting on substrates through experiments, finite element analysis (FEA) and comparison of analytical model predictions (Volkersen and Tsai) with the experimental digital image correlation (DIC) results. Experiments are carried out for double strap joint samples and three-dimensional finite element analysis is carried out with aid of ANSYS software. FEA model is used to simulate the adhesive layer's mechanical behavior, taking into account material properties and boundary conditions determined through experimental characterization. DIC data are used to understand the strain transfer mechanism of the adhesive layer in strain gauge mounting. The findings from both the FEA and experimental studies highlight the significance of the adhesive layer's properties, in obtaining precise strain measurements and strain transfer mechanisms. A thorough comparison of FEA predictions with experimental results allows for identifying critical factors that influence the accuracy of strain gauge measurements using DIC adhesive layers. This study offers guidance for choosing suitable adhesive materials and ideal mounting configurations for particular application.

Keywords adhesive layer, DIC technique, finite element analysis, strain gauge, strain transfer mechanism

1. Introduction

Different structural components are joined to create structures in aerospace, automobiles, marine, civil, and mechanical industries. This is because most structures require complex shapes and sizes, which are impossible to make with a single fabrication process. Joining different elements helps to achieve these complex shapes and sizes and also helps to achieve the strength and stability of the structure as a whole. When stress concentrations occur in joints, they can lead to a variety of issues, such as cracks, fractures, and deformation of the connected parts. This can compromise the integrity and stability of the entire structure, increasing the risk of catastrophic failure.

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Therefore, it is crucial to design and construct joints with proper reinforcement and stress distribution to ensure the overall strength and durability of the structure (Ref 1, 2). The electrical resistance strain gauges are used to measure the strain on the surface of the specimen when it is subjected to stresses on its structural members, such as complex structural surfaces. The choice of bonding material and its thickness can significantly impact the accuracy of strain gauge measurements. A mismatch in the coefficient of thermal expansion between the bonding material and the specimen can introduce errors, while a thicker bonding layer can create additional strain and distortion in the gauge, leading to inaccurate readings. The strain measurement sensor comprises a thin metallic grid, carrier material, and connections. When a strain gauge is mounted on structural components, bonding material (adhesive) can significantly reinforce the strain gauge (Ref 3-5). Several experimental studies have been conducted on bonded joint strains, particularly using strain gauges and full-field moiré interferometry (Ref 6-8). Researchers investigated strain distributions and used a high strain resolution technique, namely high-magnification Moiré interferometry, to measure the strains in the double-lap joints (DLJ), which allowed them to observe strain concentration regions in the adherends of Al/Al and Al/CFRP DLJs and, this demonstrated that, even though the adherends may behave elastically, strain concentrations can still occur in certain areas (Ref 9).

In order to optimize the structural design of aircraft structures, new trends are emerging for joining structural components composed of diverse materials. Various similar and

dissimilar materials are joined with adhesive-bonded joints in aircraft structural components. In addition to the type of adhesive and its properties, factors such as sample preparation, adhesive application method, curing conditions, and the presence of contaminants can also significantly influence the bond strength of adhesives (Ref 10). These factors must be carefully considered to ensure a strong and durable bond in the dual adhesive-bonded single-lap joint concept (Ref 11, 12). Adhesive joints offer greater structural efficiency compared to mechanically fastened joints, as they provide fewer opportunities for stress concentration elimination (Ref 13).

Choosing the right adhesive plays a crucial role in adhesively bonded joints, and its properties directly affect bond strength. The process of choosing an adhesive is typically influenced by factors such as the nature of the materials that need to be bonded, the specific temperature required for the curing process, and the intended application method. The selection of the appropriate adhesive is crucial as it directly impacts the overall performance and durability of the bonded materials, making it essential for manufacturers and engineers to carefully consider these variables before making a decision. Different types of adhesives are available, and each type provides a unique set of benefits that are particular to its application. Consequently, adhesives play a crucial role in joining structural elements in the structure for better performance. Bond strength can be improved by using a combination of different adhesives (Ref 14).

Several analytical models have been developed to analyze adhesively bonded joints (Ref 15–18). The initial contributions of researchers focused on determining stress distributions in the adhesive along the bond lines of single-lap and double-lap configurations. The researchers suggested a direct shear lag model for mechanical connections featuring numerous fasteners. Subsequently, the model was extended to incorporate adhesively bonded lap joints, assuming that the adherends experience tension while the adhesive encounters shear stress alone, both of which remain uniform throughout the thickness and also considered the influences of adherend bending, peel stress, and shear stress in the adhesive layer within a single-lap joint framework (Ref 19). (Ref 20, 21, 22) enhanced the shear lag model, while (Ref 22) proposed using a layered beam theory over the classical homogeneous beam model for single-lap joints. These modifications addressed the assumption of linear shear stress variation through the adherends in the shear lag model (Ref 23).

1.1 Double Strap Joint

Double strap joints are common in various industries like aerospace and automotive for joining different materials or repairing damaged parts. Research has shown that double strap lap adhesive joints exhibit varying strengths under static and fatigue loadings, with mismatched lap joints demonstrating longer fatigue life than matched lap joints. This improved adhesion is necessary to ensure that the joint is strong enough to withstand the weight and stress of the mega structure (Ref 24). The joint is designed with two mutually perpendicular overlapping plates, with a layer of adhesive between them. This configuration helps distribute the load across the joint evenly, and because the plates overlap, it adds strength and rigidity. If the loads applied to the joint are not balanced, the joint will only be subjected to bending (Ref 25–27). Many researchers (Ref 28–32) their studies focused on adhesive-bonded double

strap joints. Studies on compliant adhesive joints have also highlighted that aged specimen fail due to delamination within the composite. In contrast, unaged specimens fail due to cohesive damage and disbond at the adhesive-adherend interface (Ref 33). Research has also focused on the failure behavior of steel/CFRP double strap joints (DSJ) under quasi-static tensile loading. A new stress-based method, the Point Stress (PS) method, has been proposed for predicting failure loads. This method has shown good agreement with experimental results and offers a rapid and convenient approach for failure load prediction. Such advancements are crucial for improving the reliability and safety of retrofitted structures using DSJs (Ref 34). Tsai and Morton (Ref 35) conducted a study to examine the tensions present in double-lap adhesive-bonded joints that involve laminated composite adherends. Their findings revealed the uneven distribution of adhesive shear stresses attributed to variations in longitudinal stiffness. Studies have developed estimation models for bond strength and effective bond length for CFRP and corroded steel plate joints. These models, based on interfacial stress distribution and boundary conditions, have been validated against experimental data. The models provide reliable predictions and are useful for designing reinforcement systems for corroded steel structures (Ref 36). Guangyong Sun et al. (Ref 37) compared the performance of the adhesively bonded joint under different loading conditions and showed that the adhesive layer was capable of transmitting shear stresses between the two materials. He also found that the joint strength was proportional to the adhesive layer thickness and that the joint stiffness decreased with increasing adhesive layer thickness. Stigh et al. (Ref 38) assumed that the adhesive deformed elastically and that the tensile stresses developed in the adhesive layer were related to the displacement of the adherends. They then used this information to derive an equation that could be used to predict the tensile stresses in the adhesive layer.

1.2 Strain Measurement Techniques

Strain measurements are used to determine the amount of deformation a material has undergone due to an applied force. This information can provide insight into the material's strength, elasticity, and durability, which are important factors when designing and constructing mechanical components. The DIC technique offers an accurate, repeatable, and non-contact way of measuring strain in materials (Ref 39). It can also measure strains in three dimensions, making it very useful for complex shapes.

Additionally, it is used to measure strain in a wide range of materials, from metals to composites. The DIC method is a mathematical technique used to measure the displacement of a structure or object by using two or more images taken at different points in time. The study by Yoneyama (Ref 40) showed that the algorithm and basic principle of the DIC method are accurate and reliable. Additionally, the study provided a measurement example and its results to demonstrate the method's effectiveness (Ref 40). To accurately measure strain, (Ref 30) devised a two-dimensional deformation measurement algorithm. Throughout the experiment, high-resolution images were captured using a camera while testing the materials for tensile strength and examination. The algorithm was used to compare the strain measurements from the DIC and extensometry techniques to ensure accuracy. The results showed that the two techniques yielded similar results,

indicating that the DIC technique was reliable and could be used for accurate strain measurements. Comer et al. (Ref 41) concluded that the strain distributions showed significant differences between the 2D and 3D measurement techniques, indicating the importance of 3D digital image correlation for analyzing composite lap joints (Ref 41). Kashfuddoja and Ramji (Ref 42) study sought to investigate how the strain area and displacement of a one-sided corrected CFRP panel would change when subjected to a tensile load. The researchers used experimental measurements and finite element analysis to compare their results, and found that the two methods agreed (Ref 42). Shrestha et al. (Ref 43) used the DIC technique to analyze the bonding behavior of lap joints. They studied how the adhesive properties of the joints changed over different temperatures and loading conditions. They observed the Variation of the stress and strain of the joints and its effect on the overall performance of the joint. The analysis revealed that applying the silicon pollutant at higher rates caused the CFRP/Aluminum interface to be more susceptible to debonding due to decreased adhesive strength between the two materials.

Furthermore, the DIC technique was able to accurately measure the bond line strain and displacement of the joint, providing a clear indication of the effects of the silicon pollutant on the bond (Ref 43). Stuparu et al. (Ref 44) concluded that joint strength primarily depended on the adhesive used, with the adherend material playing a minor role. The specimens were tested for failure under a uniform tensile loading, and the results showed that aluminum-carbon fiber joints have high fatigue strength and good resistance to high strain rates. Cohesive zone modeling (Abaqus) proved to be an effective numerical analysis tool for predicting the lap joint strength of aluminum-carbon fiber materials. The numerical analysis of the joints revealed a good correlation between the experimental and numerical results. The DIC method allowed for a detailed analysis of the strain and deformation in the overlap length correlated with the joint strength. By comparing the results between different materials, it was found that the joint strength decreased with the increase of the strain and deformation, indicating the weakening of joint integrity and the reduction of joint rigidity.

The prediction of stress, strain, and failure patterns is important for the design of bonded joints, as it can be used to ensure that the joints are strong enough to withstand the forces they are subjected to while also ensuring that they are not overdesigned. This can help to reduce costs and improve the overall performance of the joint. The present research work is intended to study deformation, normal strain and shear strain in the overlap length of adhesive using the DIC technique. Experiments were conducted on double strap joint configuration specimens to characterize the adhesive layer subjected to the tensile shear test and to measure displacement, normal strain, shear strain, and bond strength in an adhesive layer. The algorithm for strain measurement in digital image correlation (DIC) technique utilizing Ncorr: open-source 2D digital image correlation Matlab software (Ref 45) is applied for the analysis of the captured images. Experimental measurements, analytical results and DIC method results were used to compare their results.

This paper investigates the role of the adhesive layer in mounting strain gauges on engineering structures. Accurate measurement of strain is crucial for evaluating performance and ensuring the structural integrity of these structures. Strain gauges are widely used for this purpose, and the adhesive layer is a critical component that influences the transfer of strain from the substrate to the strain gauge. This study aims to understand

the impact of variations in the adhesive layer on strain transmission under different loading conditions. To explore this phenomenon, the paper utilizes experiments, finite element analysis (FEA), and analytical models.

The study integrates experimental methods, FEA simulations, and analytical model predictions to thoroughly analyze the role of the adhesive layer in strain transfer. This multifaceted approach offers a more comprehensive understanding than studies focusing on a single method. Incorporating digital image correlation (DIC) technology to capture and analyze strain distribution adds a novel aspect. The paper compares experimental DIC results with predictions from established analytical models (Volkersen and Tsai), validating and critically evaluating these models in the context of adhesive layers in strain gauge mounting. The study includes detailed experimental setups to test double strap joint samples under various loading conditions, precise application and measurement of loads, and accurate strain measurements using strain gauges. It also involves characterizing adhesive layer and substrate materials to obtain accurate material properties for FEA simulations and interpreting experimental results. Using ANSYS software, a three-dimensional FEA model simulates the mechanical behavior of the adhesive layer, incorporating accurate material properties and boundary conditions. A DIC system captures high-resolution strain data during experiments. The study applies Volkersen and Tsai models to predict strain transfer in adhesive layers, comparing these predictions with experimental and FEA results to evaluate their accuracy. The thorough analysis of data from experiments, FEA simulations, and analytical models identifies critical factors influencing strain gauge measurements and recommends suitable adhesive materials and mounting configurations.

2. Materials and Methodology

2.1 Materials

Experimental investigations are conducted to examine the impact of adhesives on the bonding strength of joints and to ascertain the displacement, normal strain, and shear strain within the overlapped adhesive length by employing the digital image correlation (DIC) technique. The composition and mechanical properties of Aluminum alloy and adhesive are given in Table 1, respectively. Aluminum alloy plates (Al6061-T6) having a thickness of 3.2 mm, length of 108 mm, and width of 25 mm were used as substrate, aluminum alloy plates (Al 6061-T6) of thickness of 1.6 mm were used as the strap, having overlap lengths (L) of 27.4 mm. The width of straps and adherends 25.4 mm. Araldite 2015-1 (Huntsman Advanced materials) was used as an adhesive for preparation of specimen.

Table 1 Material properties of aluminum alloy and adhesive according to Campilhoet et al. (Ref 46)

Mechanical properties	Al6061 alloy	Adhesive-Araldite 2015-1
Density, kg/m ³	2700	1400
Young's Modulus, MPa	69000	1850
Poisson's ratio	0.33	0.33
Tensile Strength, MPa	260	21.63

The araldite was chosen because it provides a strong bond and is resistant to extreme temperatures and marine applications. This makes it ideal for applications that require a strong and durable bond, even in extreme weather conditions. Material properties of aluminum alloy and araldite 2015-1 in Table 1 are according to Campilho et al. (Ref 46).

2.1.1 Sample Preparation. Double-lap shear joints used for conducting experiments are shown in Fig. 1(a) and (b). It consists of two 3.2 mm thick $\times$ 25.4 mm wide and 108 mm long aluminum (Al 6061) adherends bonded together using two aluminum plates 1.6 mm thick $\times$ 25.4 mm wide and 27.4 mm long (Al 6061). At the gripping portion at two ends, knurled aluminum plates at 45° are bonded to aluminum strips using 1.52 mm thick $\times$ 25.4 mm $\times$ 31.8 mm.

The bonding surfaces of the aluminum adherend and plates underwent sandblasting and brushing with acetone to eliminate dust and soluble chemical contaminants. Two aluminum plates were then aligned end-to-end. A gap of 2 mm was left between the ends of the aluminum plates to minimize stress transfer through end-to-end bonding of the aluminum plates. Bond line thickness of 0.315 mm is observed for DSJ S1, 0.255 mm for DSJ S2, and 0.155 mm for DSJ S3. Araldite 2015-1, a combination of hardener and resin, a 400-ml dispenser is used to get resin and hardener to the ratio of 1:1. Nozzle is used to mix the proportion uniformly. The adhesive was applied to the surface of the aluminum plate using a plastic spatula to minimize the formation of air voids in the adhesive layer. The aluminum plates were positioned in alignment with the aluminum adherend, firmly secured, and left to cure at ambient temperature (30 °C) for a minimum of 24 hours prior to the removal of the clamps. Following the curing of one side of the joint, the other side of the joint was prepared utilizing an identical methodology.

2.2 Tensile test with DIC

The experimental study uses three samples to measure the bond strength between two materials in a double-lap shear configuration. The tensile test was conducted to analyze the

strength of the bond, and DSJ specimens were tested for DIC to evaluate displacement, normal strain, and shear strain in plate, adhesive, and adherend. The tensile shear test measurements were conducted at room temperature (28 °C) under displacement control mode (0.5 mm/min) using a universal testing machine, servo-hydraulic (Walter + bai), with a maximum load capacity of 25 kN. The universal testing machine was interfaced with a data acquisition system (Fig. 2).

Digital image correlation (DIC) is a widely recognized technique used to noninvasively evaluate complete surface maps of displacement and strain in a deformed specimen without physical contact (Ref 47). The digital image correlation system processes 2D analysis. The tensile tests on DSJ samples were carried out using DIC measurement methods using a servo-hydraulic machine (UTM), as shown in Fig. 3(a), with a capacity of 25 kN and a displacement rate of 0.5 mm/min. DSJ experiments were observed using one camera. The camera is used to record video placed on the side of the DSJ sample. The enlarged view of the test setup for 2D digital image correlation is shown in Fig. 3(b). Markings made on the side view of the DSJ sample vary from 0, 6.35, and 12.7 mm on the upper left and right, and 14.7, 21.05, and 27.4 mm are marked on the lower left and right of the DSJ sample, as shown in Fig. 3(b). A Sony FX3 camera with a 35 mm lens, with a maximum capacity of 120 fps, is used for video recording, as shown in Fig. 2. In this study, 100 fps was used to capture the video recording.

The experiments were videotaped and recorded at a frame rate of 1.67 frames per second (fps) from the camera. The time taken to conduct DSJ S1 is 22 sec, for DSJ S2 is 24 sec and DSJ S3 is 26 sec. A total of 37 images were taken for the DSJ S1 specimen. Forty images were extracted for the DSJ S2 specimen, and 44 images were extracted for the DSJ S3 specimen. Initially, the resolution of the photographs was 3840 $\times$ 2160 pixels. Later, the images are reframed to 448 $\times$ 1215 pixels to focus on the region of interest.

The images that are captured are subjected to analysis through the Ncoor Matlab software. This analysis employs the zero-normalized cross-correlation criterion (ZNCC) in conjunc-

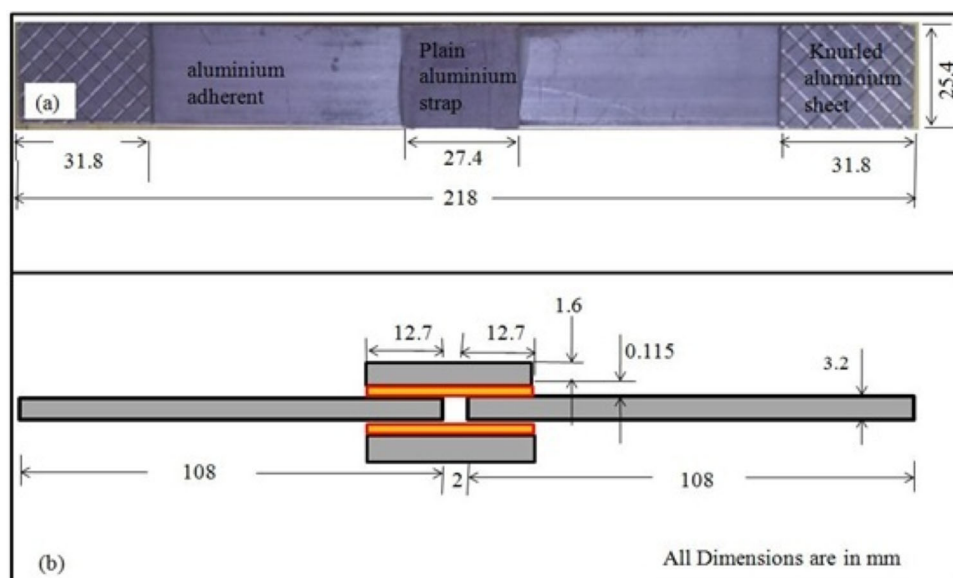


Fig. 1 (a) DSJ testing specimen dimension used for the study according to ASTM D3528-96 (Ref 50) and (b) schematic representation of DSJ specimen in cross section.

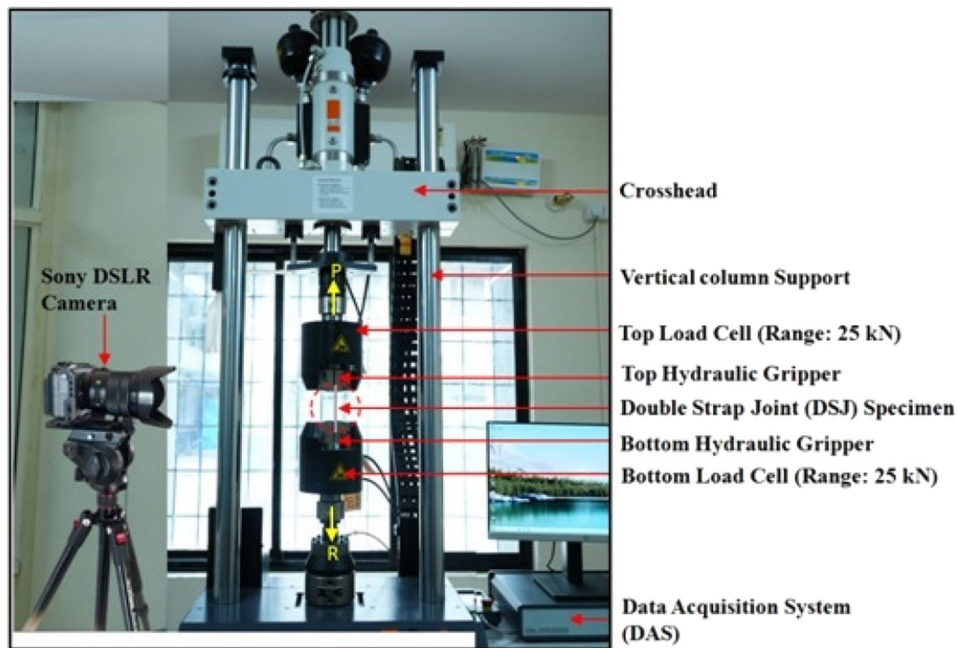


Fig. 2 Test setup for 2D digital image correlation for DSJ

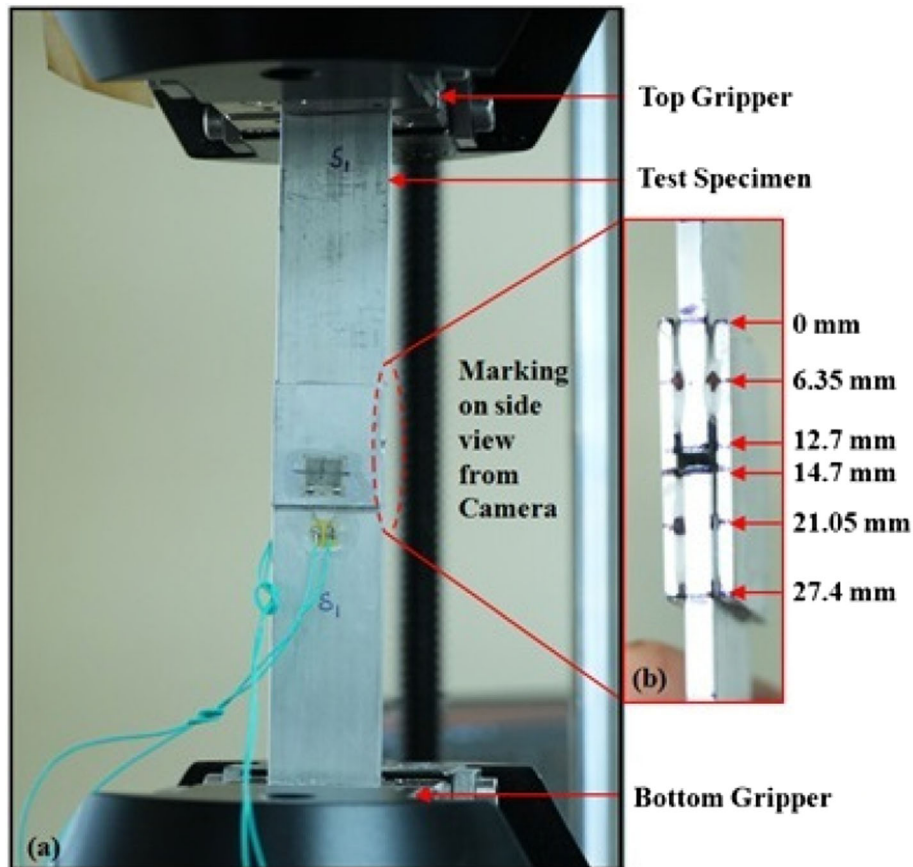


Fig. 3 (a) Enlarged view of a test setup for 2D DIC, (b) before the conduction of experiment, marking on side view from camera.

tion with the Gauss-Newton nonlinear iterative least-squares method, as (Ref 47) described. The software allows for assessing displacement and strain by focusing on a designated region of interest.

2.3 Finite Element Modeling

A nonlinear finite element module was considered to perform finite element analysis using ANSYS software on

double strap joint samples. The mechanical properties of adherend (AL6061-T6), adhesive, and Strap (AL6061-T6) used in constructing the nonlinear finite element model are given in Table 1.

The modeling and meshing of double strap joint samples utilized SOLID185 elements for analyzing deformation along the X, Y, and Z axes. This element type is a higher-order three-dimensional solid element exhibiting quadratic displacement behavior, featuring 10 nodes with three degrees of freedom per node. It consists of 7 nodes on the outer and inner surfaces, along with 3 nodes in the mid-thickness of the model. Meshing criteria were set to be fine, with shape transition and span angle center considered medium, and the element area for DSJ meshing was 0.3175 mm. Higher-order formulations like 6 node triangles and 8 noded rectangles were deemed more accurate for triangle and rectangle elements. The adherend, adhesive, and strap regions were defined separately and divided into elements, with smaller mesh sizes used in critical strain distribution zones. The material model employed was multi-linear isotropic hardening von Mises plasticity (MISO), with strap analyses considered elastic and plate (AL6061-T6) and adhesive analyses considered elastoplastic. Mesh details of double strap joint finite element models are given in Table 2. The meshed double strap joint sample is shown in Fig. 4(a).

Figure 4 illustrates the double strap joint model schematic, which elucidates the boundary conditions utilized in the finite element simulation. A tensile load was applied on one end of the adherend while the other end of the specimen was fixed where 'P' represents applied load and 'R' represents reaction at the fixed end of the adherends as shown in Fig. 4(b).

3. Results and Discussion

Experiments are conducted on double strap joint (DSJ) specimens subjected to tensile forces and adhesive surface experiences tensile shear force, and the results obtained are presented. In addition, digital image correlation (DIC) and numerical analysis were conducted, and the results that were obtained were showcased. A comprehensive comparison and discussion between the experimental and numerical findings are presented in detail.

3.1 Load-Displacement Response of Double Strap Joint Samples

The experimental study is carried out on three samples to measure the bond strength and strain transfer mechanism between two materials in double-lap strap joint configurations. Tensile testing utilizes a universal testing machine to examine the behavior of materials under tension. The resulting load versus displacement curves for three distinct specimens are depicted in Fig. 5. The experiment results obtained for double strap joint models are given in Table 3.

Table 2 Mesh details of double strap joint finite element models

Model	No. of total nodes	No. of total elements
Double Strap Joint	744186	149816

Table 3 shows that DSJ S1 exhibited a slightly lower failure load of 1.3 kN and a corresponding shear stress of 2.02 MPa in double strap joint samples. DSJ S2 displayed similar behavior to DSJ S1, with a moderate failure load of 1.75 kN and a related shear stress of 2.71 MPa. DSJ S3 demonstrated the highest failure load among the three samples, with a load of 2.45 kN and a shear stress of 3.79 MPa.

In Fig. 5, the load-displacement response of the DSJ specimens (DSJ S1 to DSJ S3) exhibits a linear trend from the onset of testing until ultimate failure. Based on the load vs displacement behavior of the double strap joint (DSJ) samples, the adhesive appears to exhibit a brittle adhesive behavior. Notably, there is no substantial presence of plasticity leading to brittle fracture. Initially, the load value rises linearly with displacement, indicating elastic behavior of the materials. DSJ S1 sample followed a similar trend to DSJ S3, but with a lower maximum load. The load reached around 1.3 kN at a displacement of 0.1122 mm before dropping sharply. The performance of DSJ S1 was similar to DSJ S2, though the failure occurred earlier and at a lower load, suggesting weaker joint performance or earlier failure due to joint configuration. The mode of failure observed in DSJ S1 is adhesive, maximum adhesive retained on both sides (plate and adherent), as shown in Fig. 6(a). DSJ S2 sample exhibited a slightly higher load capacity compared to DSJS1, peaking at around 1.75 kN at a displacement of 0.1246 mm. DSJ S2 had a weaker performance than DSJS3. The drop in load after reaching maximum suggests that the joint might have experienced debonding or material failure. The mode of failure observed in DSJ S2 is an adhesive failure, as shown in Fig. 6(b). Microcracks develop within the deformed area, leading to both localized and gradual structural breakdown. This localized and progressive deterioration precedes a catastrophic failure, ultimately defining the failure mode of the joint. For DSJ S3 sample, the load increased steadily with displacement, reaching a maximum load of 2.45 kN at a displacement of 0.1528 mm. The load vs displacement curve indicates consistent elastic behavior up to this point, with no significant drop in load, suggesting the joint maintained its structural integrity throughout the test. DSJ S3 sample demonstrates a stable load-bearing capacity without sudden failure, indicating good bonding and strength in the double strap joint. The mode of failure observed in DSJ S3 is cohesive failure, where adhesive is retained maximum on the plate compared to adherent, as shown in Fig. 6(c). It is observed that initially, the adherends deform, later cracks start developing in the bonded region, and the adhesive material softens, leading to catastrophic failure. DSJS3 showed the highest load-bearing capacity and stable behavior throughout the displacement range, indicating the strongest joint among the three samples. The linear increase in the load-displacement curves for all models indicates that the joint exhibits some degree of elasticity and stiffness, as it responds to proportional changes in load under small deformations. DSJS1 and DSJS2 showed signs of early failure, with sharp drops in load after reaching their peak load, indicating weaker joint bonding or premature failure compared to DSJS1. The sharp drops in load for DSJS1 and DSJS2 suggest possible debonding or joint failure, while DSJ S3 exhibited more gradual behavior without such signs. The differences in performance between the samples could be attributed to variations in material properties, bonding quality, or even slight differences in specimen preparation. DSJ S3 may have had better bonding or material consistency, leading to its

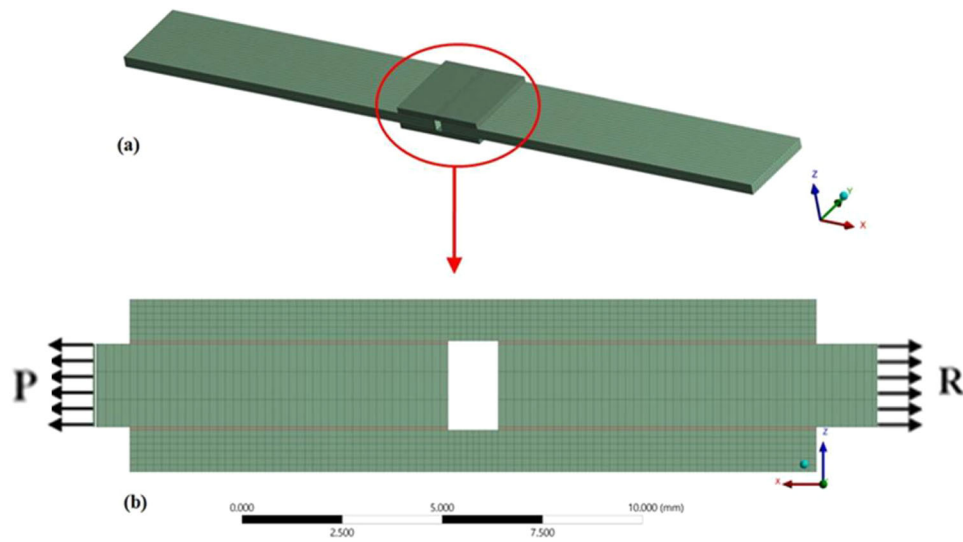


Fig. 4 The meshed model of the double-bonded strap joint (a) whole cross-section (b) enlarged view of meshing is considered in the strap, adhesive, and adherend.

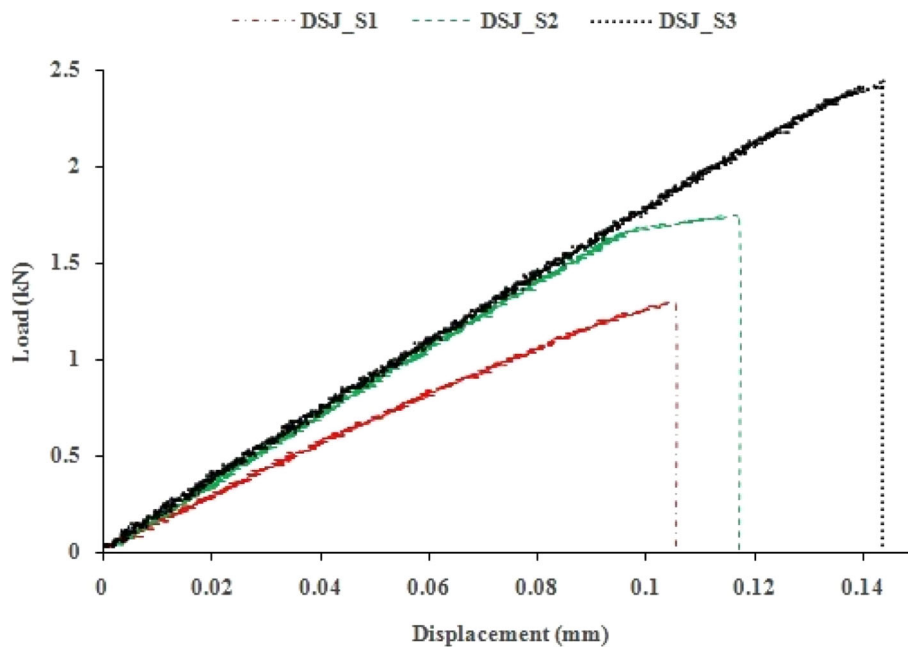


Fig. 5 Load versus displacement curves for double-lap strap joint samples.

Table 3 Experiment results for double strap joint models

Double strap joint samples	Adhesive layer thickness, mm	Failure load, kN	Area, mm ²	$\tau = P/2A$, MPa
DSJ S1	0.315	1.3	12.7 × 25.4 mm	2.02
DSJ S2	0.255	1.75	12.7 × 25.4 mm	2.71
DSJ S3	0.155	2.45	12.7 × 25.4 mm	3.79

superior performance. The early failures in DSJ S1 and DSJ S2 point toward deficiencies in these areas.

In the case of dual adhesive-bonded joints as shown in Fig. 7(a) for one of the lap joint, it's essential to emphasize that the separation at the interfaces among the plate, adhesive, and

adherends is intricate. Through the current investigation, it has been uncovered that the adhesion mechanism predominantly relies on two mechanical interlocks, initially occurring at the interface between the adherend and adhesive, or at the interface between adhesive and plate. To better understand this, consider

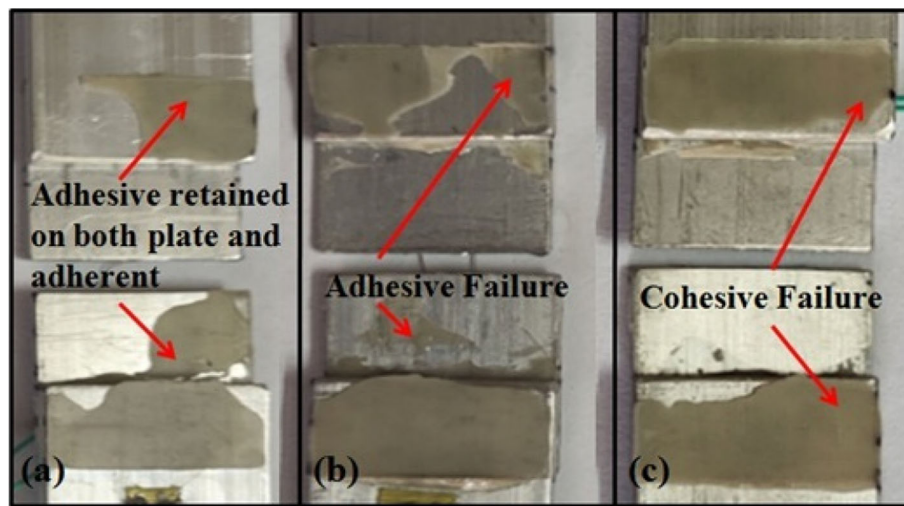


Fig. 6 Photographs of dual adhesive bond joints after testing

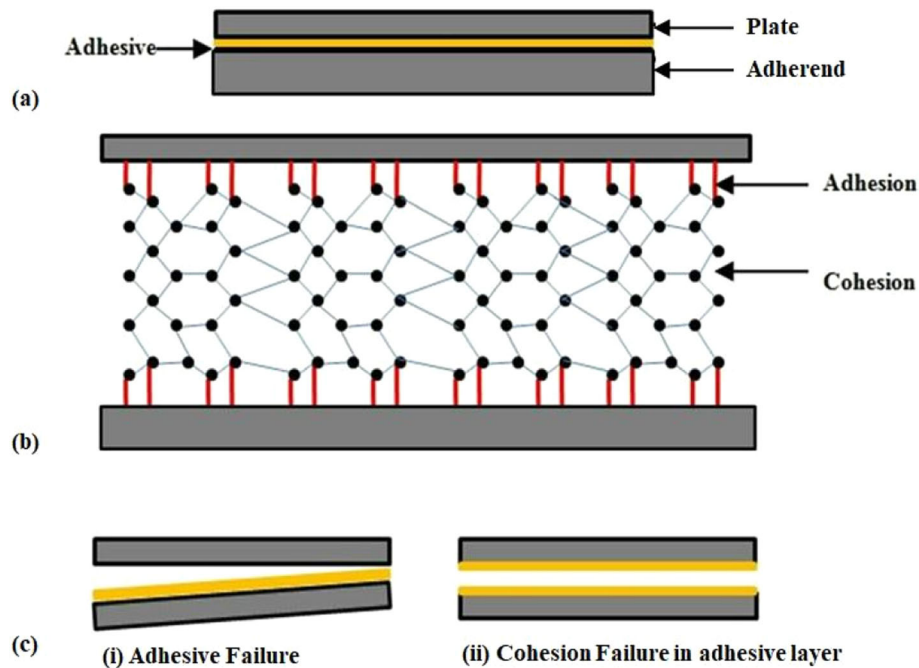


Fig. 7 Hypothetical illustration of adhesion bond (a) DSJ model in one of the lap joint (b) molecular interaction b/w adherend and plate (c) modes of failure

the hypothetical adhesion bond shown in Fig. 7(b). Mechanical interlocking relies heavily on the surface roughness created for the bonding process. Essentially, the adhesive enters into the pores and other irregularities on the adherend's surface, forming a mechanical lock between the adherends and plates. In this context, adhesion occurs predominantly through this mechanical interlocking at the interface between the aluminum adherend, Araldite adhesive, and plate. The interaction between the aluminum adherend, Araldite adhesive, and aluminum plate involves concurrent adhesion mechanisms. Simultaneously, a similar interplay occurs between the aluminum adherends themselves. This intricate combination of mechanical interlocking and diffusion forms the foundation of our research focus, especially when a member is loaded, failure mode

(Fig. 7c) and how it gets transferred through an adhesive medium. The non-contact mechanism used to study this complex interrelationship is through digital image correlation.

Figure 8 illustrates the forces acting on a Double strap Joint specimen subjected to tensile shear force and critical points used for DIC.

Figure 5 shows the specimens DSJ S1, DSJ S2, and DSJ S3 show linear behavior. For the digital image correlation (DIC) analysis, the decision to select the DSJ S3 specimen was due to its cohesive mode of failure. In the case of DSJ S3, the adhesive shows signs of cohesive failure, indicating that the adhesive material failed in a controlled manner rather than debonding from the adherend. This is often desirable in structural applications, as cohesive failure suggests that the adhesive's

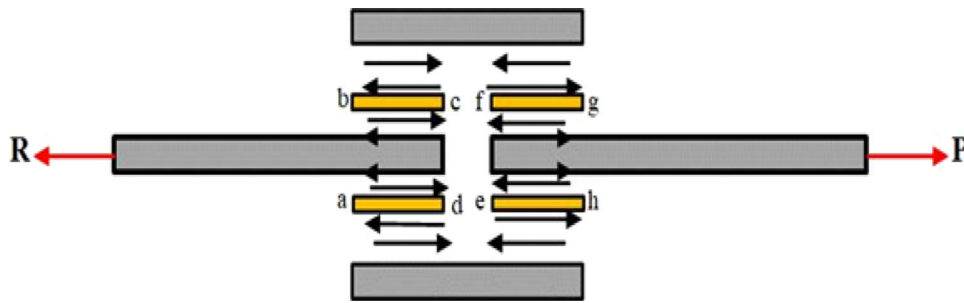


Fig. 8 Free body diagram of double strap joint sample

full strength was utilized before failure. DSJ S3 exhibited superior load-bearing capacity, DSJ S3's behavior provides critical insight into the joint's failure mechanisms. Cohesive failure often reflects a more consistent and predictable fracture process within the adhesive layer or material itself, making it an ideal for detailed strain and displacement analysis using DIC. This selection allows for a more precise observation of the material's deformation and strain distribution leading up to and during failure. Digital image correlation (DIC) will capture the surface displacement patterns and provide a visual understanding of the progressive damage. This cohesive failure mode is crucial for evaluating performance improvements in future joint designs.

3.2 Digital Image Correlation (DIC) Results

Tensile tests of DSJ samples are recorded. The digital image correlation (DIC) measurement method is used to investigate surface strain, especially in the zero strain point range. The displacement and strain values obtained using the DIC method, along and across the adhesive joint thickness, are represented in this section. DIC is evaluated for plate (very close to the adhesive), adhesive, and adherend (very close to the adhesive) on the upper left and right where the tensile load acts and downwards left and right where DSJ specimens are subjected to reaction.

3.2.1 Variation of Displacement Along the Bond Length.

The displacement of double strap joint samples is measured along the y-axis, i.e., from upper to lower DSJ joint along the bond length, using the DIC method. The displacement Contours obtained for loads of 1.0 and 2.45 kN is shown in Fig. 9(a) and (b). Variation of displacement from upper to lower of the bond length in the case of plate, adhesive, and adherend layer for a load of 1.0 and 2.45 kN is shown in Figures 10, 11, 12 and 13.

Based on the observations in Figs. 10 and 11, it is observed that there is a general trend of displacement increasing on both the left and right sides of the plate as the bond length approaches the mid-bond region (around pixels 150-160), followed by a gradual reduction as the bond length increases toward its maximum. The adhesive shows more significant displacement variations compared to the plate, with the highest displacements around the middle of the bond length, as expected. The adhesive layer transfers the stress between the plate and adherend, leading to more deformation under load. The adherend, like the adhesive, shows higher displacements, especially in the mid-bond region, reflecting its deformability under the applied load. The displacement trends in the adherend are consistent across both sides, indicating a balanced stress distribution. The right side of the plate generally shows slightly

higher displacement values than the left side across the bond length. This could be due to slight asymmetry in the stress distribution or material properties. Despite the slight differences, the displacement variations are minimal on both sides, suggesting that the plate layer is relatively rigid, resisting deformation effectively. The adhesive exhibits similar displacement trends on both sides, with peaks occurring around the mid-bond region. However, the right side shows slightly higher maximum values, especially around 160 pixels. The adhesive layer's primary role is to bond the adherend and plate, and as such, it is expected to undergo more deformation than the plate. The nearly symmetrical displacement on both sides indicates a uniform stress transfer. The displacement in the adherend layer is also comparable between the left and right sides. There are slight variations in the magnitude, with the right side showing marginally higher displacement values at some points. The adherend layer, being part of the load-bearing system, deforms significantly under stress, especially in the middle region of the bond length. The mid-bond region (around 150-160 pixels) shows the highest displacement for all layers (plate, adhesive, and adherend) on both sides. This is expected, as the middle region of the bond experiences the highest shear and tensile stresses under the applied load. Displacement values generally decrease as the bond length extends beyond the mid-region, indicating that the stress and corresponding deformation reduce toward the edges of the bond. The adherend and adhesive layers consistently show higher displacements than the plate, which is a typical behavior since the plate is more rigid and the adhesive and adherend accommodate more deformation. The plate is likely made of a stiffer material, so it exhibits the least displacement. The adhesive, being more elastic, shows the highest variation in displacement. The adherend is somewhat in between, as it deforms more than the plate but less than the adhesive. The adhesive is designed to deform and transfer stress, making its higher displacement values expected. The adherend, likely being a structural component, experiences significant deformation but remains within reasonable limits compared to the adhesive. The uniform displacement trend across the left and right sides suggests that the load distribution is symmetrical, with the middle region of the bond length undergoing the highest stress. This stress concentration causes the peak displacement observed around the middle bond length. The slightly higher displacement values on the right side for all layers may indicate a minor asymmetry in load application or material property variation between the left and right sides. Yet, the differences are small, suggesting that the overall stress distribution is quite uniform. Figures 12 and 13 show that the plate shows a peak displacement in the middle bond length (around 100-110 pixels) on both sides. Displacement decreases

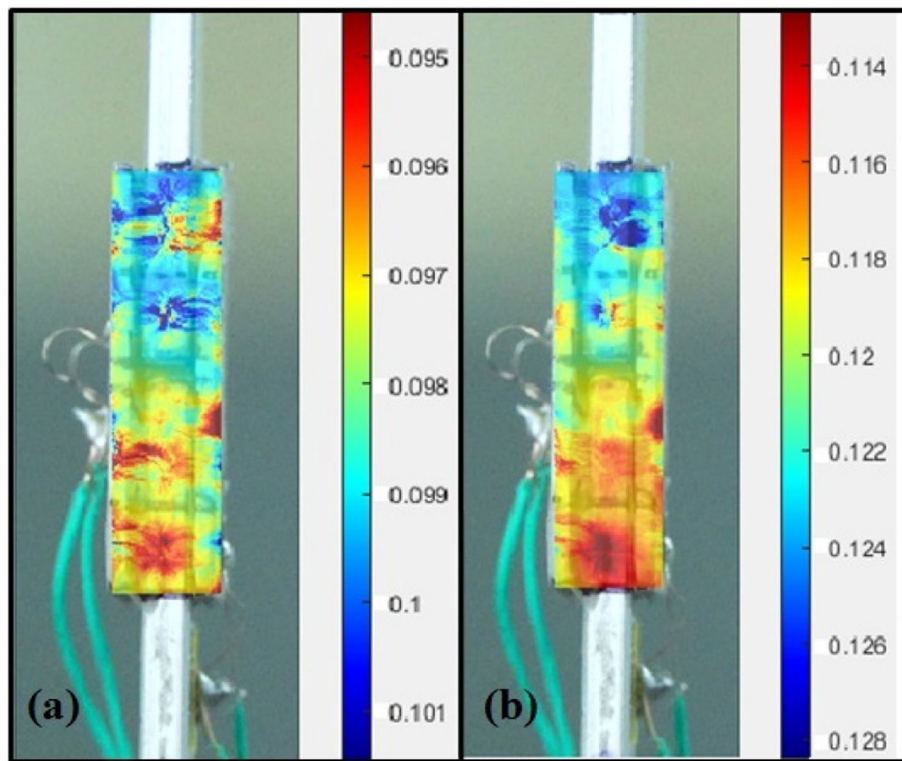


Fig. 9 Displacement contours along the DSJ for sample S1 for the load of 1.0 kN (elastic load) and 2.45 kN (failure load) using Ncoor software

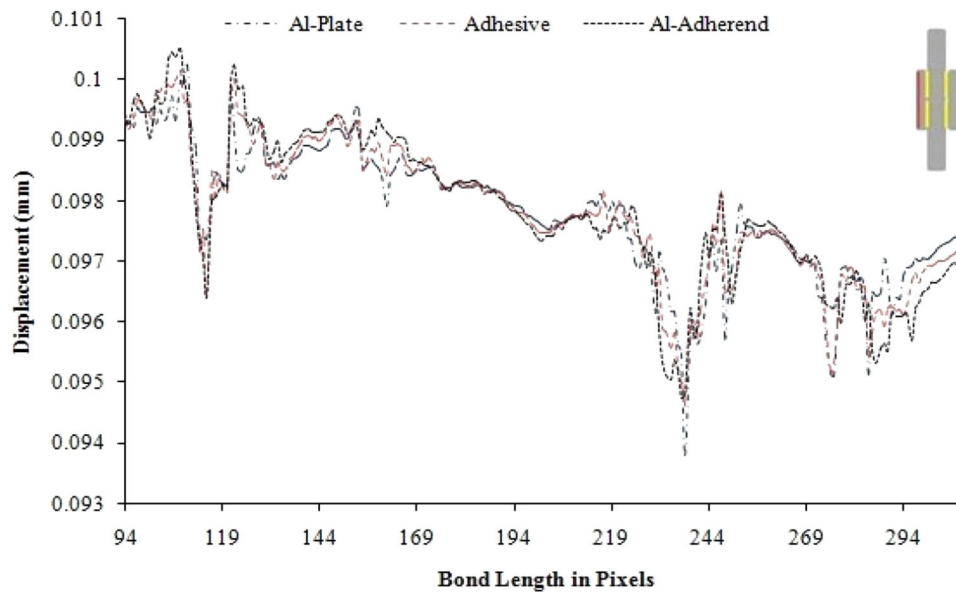


Fig. 10 Variation of displacement along the left side of bond length in case of plate, adhesive, and adherend layer for the load of 1.0 kN (elastic load)

toward the edges of the bond length. This behavior suggests a concentration of stress near the center of the bond length, which is typical under load. The adhesive experiences a noticeable peak displacement around the middle of the bond length. The higher values on the right side (up to 0.1339 mm) reflect the adhesive's significant role in absorbing the stresses, especially at failure loads. The adhesive deformation is more pronounced in the middle, consistent with its role in transferring load

between the plate and adherend. The adherend shows considerable deformation, particularly in the middle of the bond length. Like the adhesive, the right side shows slightly higher displacements, suggesting asymmetry in the stress distribution at failure load. However, the general trend is consistent with peak displacement in the middle and reduced values near the edges. Displacements on the right side of the plate are generally slightly higher than on the left, with the peak displacement at

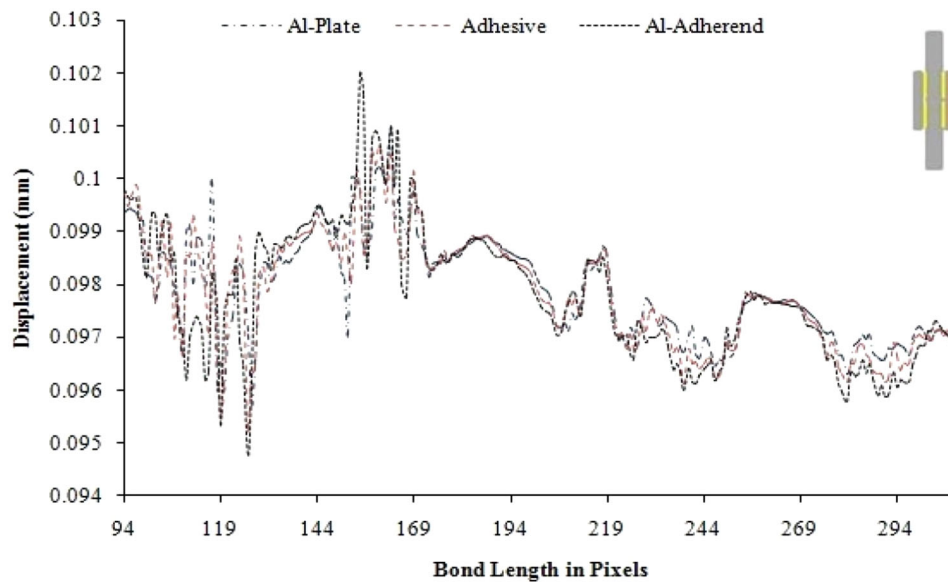


Fig. 11 Variation of displacement along the right side of bond length in case of plate, adhesive, and adherend layer for the load of 1.0 kN (elastic load)

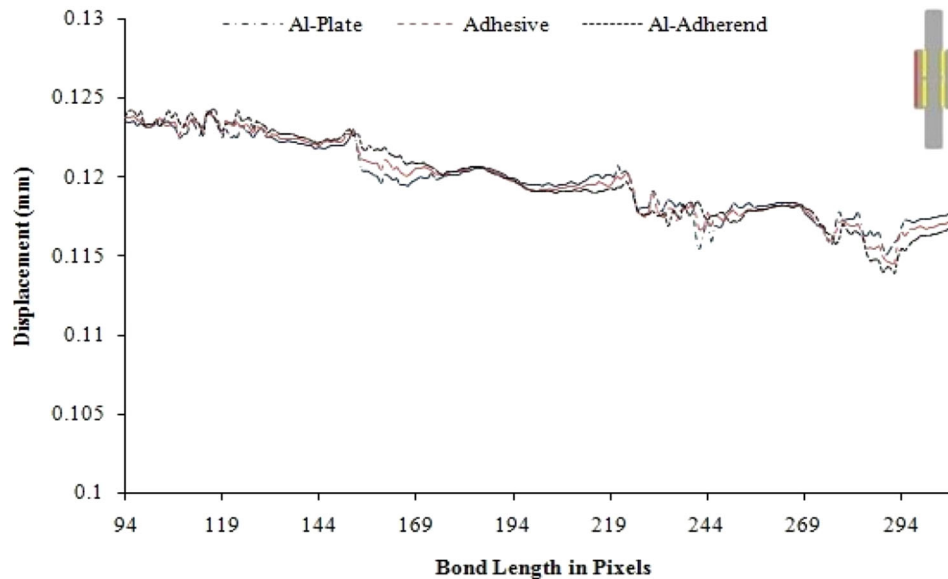


Fig. 12 Variation of displacement along the left side of bond length in case of plate, adhesive, and adherend layer for the load of 2.45 kN (failure load)

0.12807 mm compared to 0.12358 mm on the left side. This suggests that the right side of the plate is experiencing slightly more deformation under the applied load. The symmetry is relatively well-maintained, but this difference indicates a potential slight misalignment or non-uniform load distribution during testing. The adhesive shows significant deformation, particularly on the right side, where the peak displacement is 0.13390 mm (at 114 pixels), compared to 0.12421 mm on the left side. The right side consistently shows higher displacements than the left, indicating a potential uneven stress concentration or variation in material properties. This asymmetry could be the result of local bond failure or imperfections in the adhesive layer. The adhesive's higher deformation, especially in the mid-region, reflects its role in stress transfer and susceptibility to failure under high load. The adherend

follows a similar pattern, with the right side showing higher peak displacement (0.13367 mm at 113 pixels) than the left (0.12421 mm at 105 pixels). The adherend layer shows the highest displacements among the layers, suggesting it is experiencing the most significant deformation, likely due to its position in the joint structure and load-bearing role. Though present, the difference between the left and right sides is relatively consistent with the trends observed in the adhesive layer. The mid-bond region (around 100-120 pixels) shows the highest displacement for all layers (plate, adhesive, and adherend) on both the left and right sides. This is the region where the joint likely experiences the highest shear and tensile stresses, leading to the most significant deformation. Displacement decreases toward the edges of the bond length, particularly beyond 200 pixels, suggesting that the stress concentration

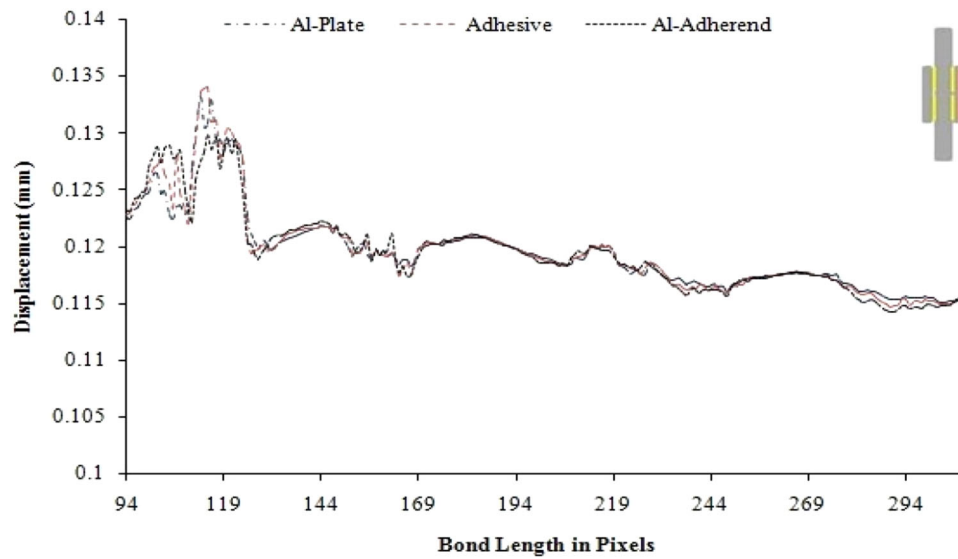


Fig. 13 Variation of displacement along the right side of bond length in case of plate, adhesive, and adherend layer for the load of 2.45 kN (failure load)

decreases away from the center. This is typical for double-shear joints, where the maximum load is borne by the middle of the joint. The adhesive and adherend layers exhibit higher displacement compared to the plate, which is expected due to the adhesive's flexibility and the adherend's role in bearing the load transferred from the plate. The plate is a stiffer material, explaining its lower displacement than the other layers. It shows the least deformation because it resists the applied load more effectively. The adhesive layer is more elastic and shows significant deformation. Its role in stress transfer between the plate and adherend makes it more susceptible to higher displacement, especially near failure load. The adherend, being a load-bearing component, shows the highest displacement, reflecting its role in absorbing the stress from the joint. The material properties of the adherend allow it to deform under high load, but its displacement is still higher than the plate. The peak displacement in the mid-bond region is consistent with the distribution of shear and tensile stresses in the joint. The stress concentration is highest near the middle of the bond length, leading to maximum deformation in that region. The right side consistently shows slightly higher displacements across all layers. This asymmetry could be due to minor variations in material properties, load misalignment, or bonding imperfections. However, the overall distribution remains relatively balanced, with deformation decreasing toward the edges. The stabilization or slight decrease in displacement at longer bond lengths might indicate that the DSJ Specimen reaches a point of equilibrium or structural resistance.

The more significant displacement occurs in the loading direction (acting upwards), and the aluminum adherend deforms more. The fluctuations in displacements (Figures 10, 11, 12 and 13) occur in the bonded region due to bending moments (eccentric loading) which form the normal and shear strains; thus, these strain values assist in detaching the adherends in the bonded region.

3.2.2 Variation of Normal Strain Along the Bond Length (ϵ_{yy}). The normal strain from the upper to lower adhesive joint, i.e., along the longitudinal axis, is measured using the

DIC method. The strain values obtained along the bond length for 1.0 and 2.45 kN are shown in Fig. 14(a) and (b). Variation of normal strain along the bond length in case of plate, adhesive, and adherend layer for the load of 1.0 and 2.45 kN is shown in Figs. 15, 16, 17 and 18.

Figures 15 and 16 provide insight into the variations in normal strains within the three layers—plate, adhesive, and adherend—along the bond length, ranging from pixel 96 to pixel 310. These strains can be positive (stretching) and negative (compression) at different points. The left side exhibits a more prominent compressive strain at the beginning (around -0.0024), while the right side generally starts in tension. This indicates that there may be a slight asymmetry in the load is applied or the materials respond to stress. Peak tensile strain occurs in the mid-bond region on both sides, with higher strain values observed in the adhesive and adherend layers compared to the plate. The adhesive shows the highest strain, reflecting its critical role in transferring load between the plate and adherend. The strain values near the edges of the bond tend to return to compressive strain, indicating that the load is concentrated in the middle of the joint, with less deformation near the bond's edges. Around 100-130 pixels this region shows the highest tensile strain for all layers (plate, adhesive, and adherend) on both the left and right sides. This suggests that this region's highest stress is concentrated, as expected in a double-shear joint under load. Beyond 230 pixels, the strain values in the plate, adhesive, and adherend transition to compression near the bond's edges. This is consistent with the fact that the joint experiences lower stress in the outer regions. The left side generally starts in compression, while the right side begins in tension for all layers. This may be due to slight misalignment or differences in the bonding process, but overall, the pattern of tensile strain in the middle and compressive strain at the edges is consistent. The adhesive shows the highest strain values, reflecting its critical role in distributing stress between the plate and adherend. The large variation in strain, from tensile to compressive, indicates the adhesive's ability to deform and accommodate the load. The adherend also exhibits high strain, indicating that it is the primary load-bearing material in the

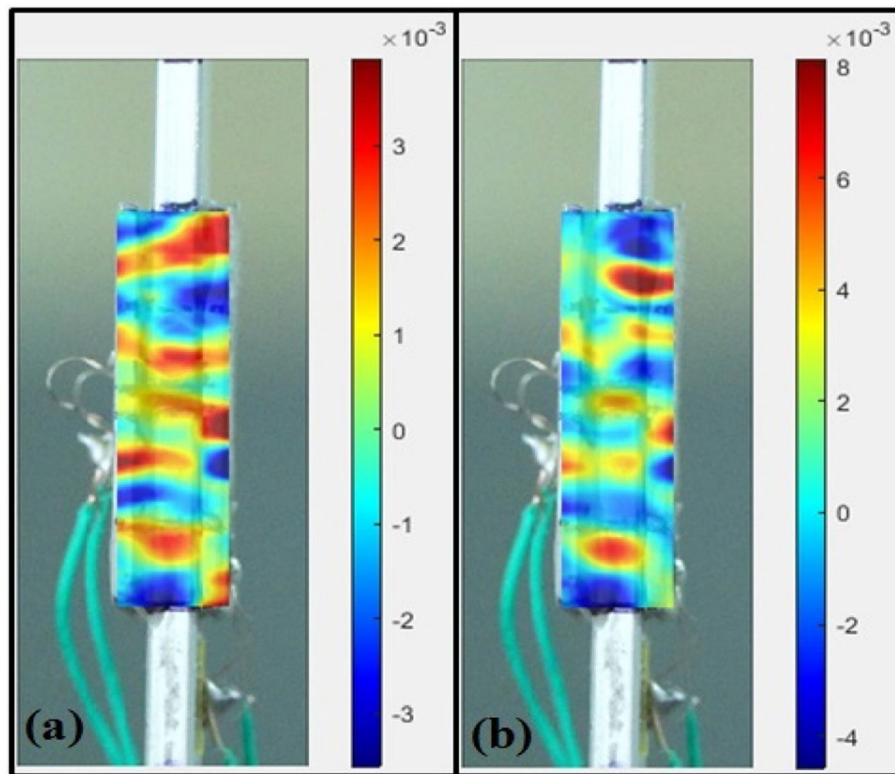


Fig.14 Normal strain ε_{yy} contours along the DSJ for sample S1 for the load of 1.0 kN (elastic load) and 2.45 kN (failure load) using Ncoor software

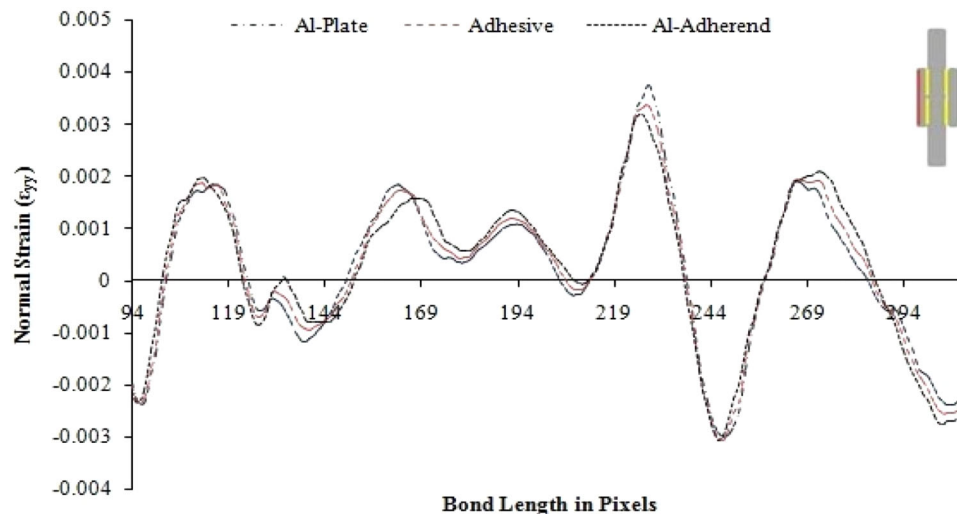


Fig. 15 Variation of normal strain along the left side of bond length in case of plate, adhesive and adherend layer for load of 1.0 kN (elastic load)

joint. Its strain behavior is influenced heavily by the adhesive's performance. The plate, being stiffer, shows lower strain values than the adhesive and adherend, but its strain behavior still reflects the general stress distribution pattern along the bond. Figures 17 and 18 show that the strain values fluctuate between positive (tensile strain) and negative (compressive strain), showing deformation of the materials due to applied load. The left side consistently shows tensile strain, indicating that left side sustains more tension, while the right side undergoes significant compression near the load application point (early

bond length). The transition to tensile strain on the right side in the mid-to-end region reflects the joint's attempt to balance the load distribution, with the plate stretching more at the center. The adhesive layer is critical for transferring stress between the plate and the adherend. The observed tensile strain on the left side and the eventual transition to tension on the right side reflect the adhesive's role in distributing the load. This behavior highlights the non-uniform stress distribution, where the adhesive deforms to accommodate the shifting stress, especially as the load increases toward failure. The adherend layer is the

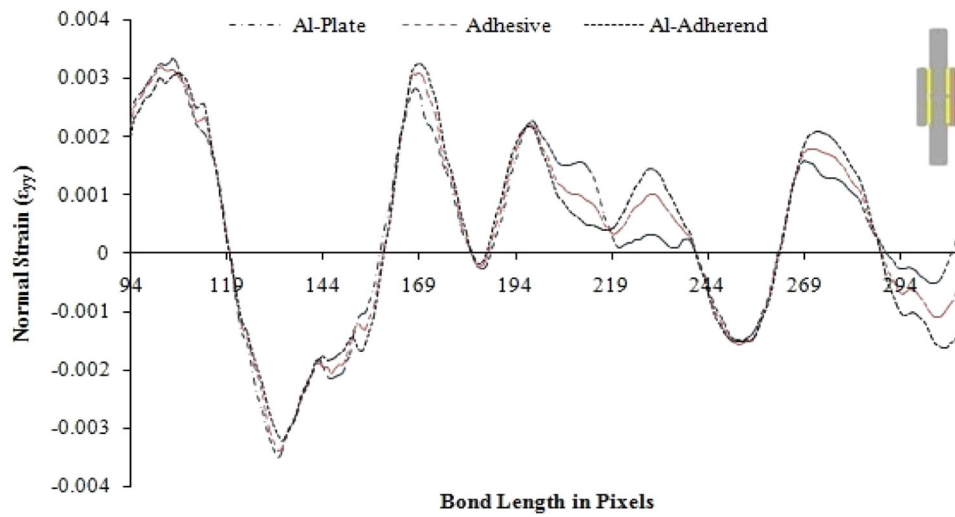


Fig. 16 Variation of normal strain along the right side of bond length in case of plate, adhesive and adherend layer for load of 1.0 kN (elastic load)

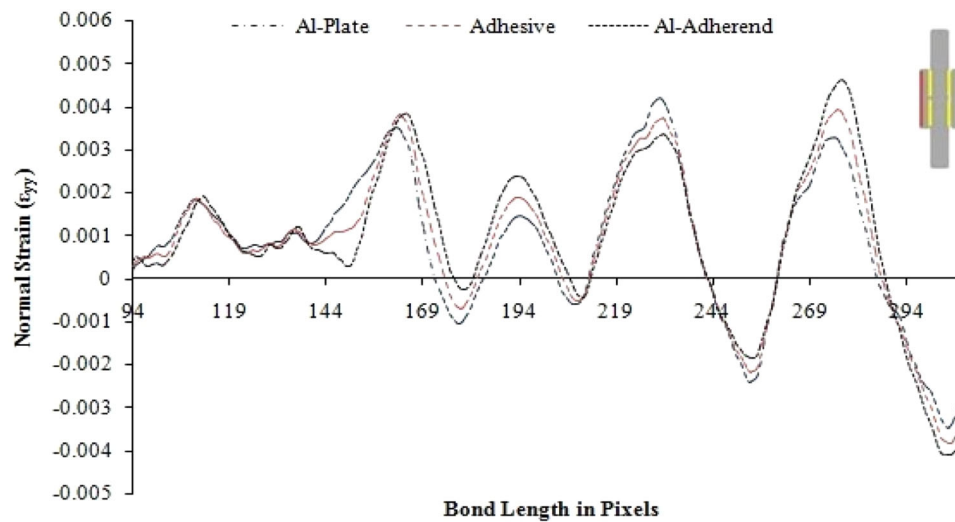


Fig. 17 Variation of normal strain along the left side of bond length in case of plate, adhesive and adherend layer for load of 2.45 kN (failure load)

primary load-bearing element in the joint. The strain in the adherend matches the adhesive behavior, reflecting the transfer of stress through the bond. The gradual increase in tensile strain on both sides, especially in the mid-bond region, indicates that this part of the bond experiences the highest stress, where the adherend stretches the most under the failure load. The left side generally exhibits tensile strain across the entire bond length for all layers (plate, adhesive, adherend), indicating that this side experiences more elongation as the load increases. The right side shows compressive strain in the early bond length for all layers, transitioning to tensile strain after 200 pixels, particularly for the adhesive and adherend. This suggests that the right side undergoes more compression near the load application point and elongation further along the bond. In the mid-bond region (200-230 pixels), most layers on the right-side transition from compression to tension, highlighting that this region bears the highest stress under a 2.45 kN failure load. On the left side, this region continues to experience increasing tensile strain, reflecting the joint's response to the applied load. The left side

consistently experiences tensile strain, which may indicate that this side is loaded more directly in tension, causing it to stretch more evenly across the bond. The right side, however, undergoes significant compressive strain in the early bond length, suggesting that this side is more constrained or loaded differently, causing it to bear more compression near the load application. This asymmetric strain distribution could result from slight misalignments or variations in how the load is applied or distributed across the joint. It could also be due to material property differences between the layers, particularly in the adhesive. This region bears the highest tensile strain on both sides, particularly in the adhesive and adherend layers. This indicates that the mid-bond region experiences the highest stress under the applied load, which is consistent with the failure load conditions. The adhesive shows the largest strain values on both sides, indicating that it undergoes significant deformation as it transfers load between the plate and adherend. This suggests that the adhesive plays a critical role in accommodating the joint's response to high loads, especially

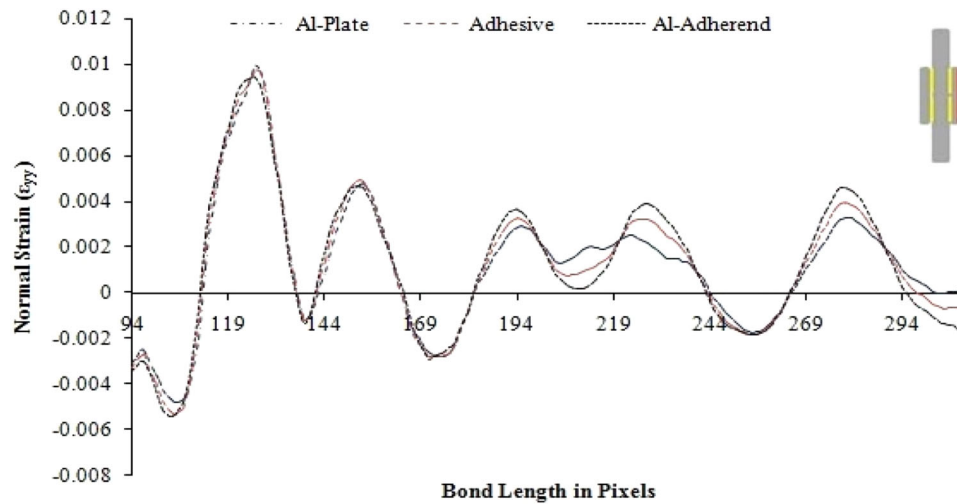


Fig. 18 Variation of normal strain along the right side of bond length in case of plate, adhesive and adherend layer for load of 2.45 kN (failure load)

near failure. The adherend shows a similar trend, with significant tensile strain, particularly in the mid-bond region, indicating that it stretches under load. The plate exhibits lower strain values compared to the adhesive and adherend, reflecting its stiffer nature. However, the transition from compression to tension on the right side indicates that the plate also deforms to accommodate the load. The plate also shows its maximum tensile strain in this region, reflecting the overall deformation of the joint under load. Beyond 250 pixels, Strain values start to decrease near the edges of the bond, particularly on the right side, where the strain transitions back toward compression after peaking. This indicates that the stress is concentrated in the middle, with less stress near the bond's ends. On the right side, the bond length up to 150 pixels experiences compressive strain across all layers. This is likely due to the loading mechanism, where the right side is constrained initially, resulting in compressive stresses near the load application point. Figure 14(a) and (b) shows the normal strain distributions along the y-axis obtained using increasing force from the middle of the adhesive layer in elastic limit and up to failure load. The fluctuations in normal strains along the adhesive bond length are presented in Figs. 15, 16, 17 and 18. Observations revealed that strain values increased with increasing load, peaking at the midpoint of the overlap region and decreased toward its ends. Additionally, it was noted that these deformations, initially compression strains in the middle of the overlap, transitioned into tension strains as the load approached the critical damage threshold with further load increments.

3.2.3 Variation of Shear Strain Along the Bond Length (ϵ_{xy}). The strain from the upper to lower adhesive joint, i.e., across the longitudinal axis, is measured using the DIC method. The strain values obtained along the plane of shear for 1.0 and 2.45 kN are shown in Figs. 19(a) and (b). Variation of normal strain along the bond length in the case of plate, adhesive, and adherend layer for the load of 1.0 and 2.45 kN is shown in Figs. 20, 21, 22 and 23.

From Figs. 20 and 21, on the left side, the strain values are predominantly negative, indicating shear strain in compression at the bond length's beginning, but it gradually transitions to positive values along the length. On the right side, the values are mostly positive, indicating shear strain in tension, which

increases steadily along the bond. The shear strain in the plate increases along the bond on both sides, but the right side starts in tension, while the left side transitions from compression to tension as the bond length increases. This is typical in lap joints, where the strain distribution varies depending on the load application and how the joint is constrained. The rise in tensile strain along the bond suggests that the plate is deforming to accommodate the applied lateral forces. The adhesive plays a critical role in transferring the load between the plate and adherend. The compression to tension transition observed on the left side reflects the adhesive's behavior under a relatively uniform load distribution. The right side shows consistent tensile strain development, indicating that the adhesive is being stretched more evenly along this side, which is typical when the adhesive must accommodate lateral forces to prevent debonding or failure. The adherend supports the load transfer from the adhesive and plate, showing a similar progression of shear strain from compression to tension. The consistent rise in tensile shear strain along both sides indicates that the adherend stretches significantly, accommodating the lateral forces imposed by the 1.0 kN load. The asymmetry in shear strain distribution between the left and right sides is typical in double-shear joints where the load is not perfectly symmetric. The left side undergoes compression at first, while the right side is loaded in tension, reflecting the non-uniform stress distribution across the bond. The transition to tensile strain on the left side indicates that the load is gradually shared more evenly as it is transferred through the joint. For bond length 150 to 200 pixels, the shear strain in all layers on both sides increases gradually in the mid-bond region, reflecting the joint's ability to distribute the load along the bond length. The adhesive and adherend layers show relatively higher strain levels here, suggesting that this region bears the most significant shear deformation. At the bond's end length of 250 to 310 pixels, the shear strain values for all layers converge to their maximum values, indicating that this region bears the most significant tensile shear strain under the applied load. The values for both sides are quite similar by the end of the bond length, suggesting that the load is more evenly distributed at this point. The left side of the bond experiences compressive shear strain at the beginning but transitions to tensile strain along the bond length. This pattern

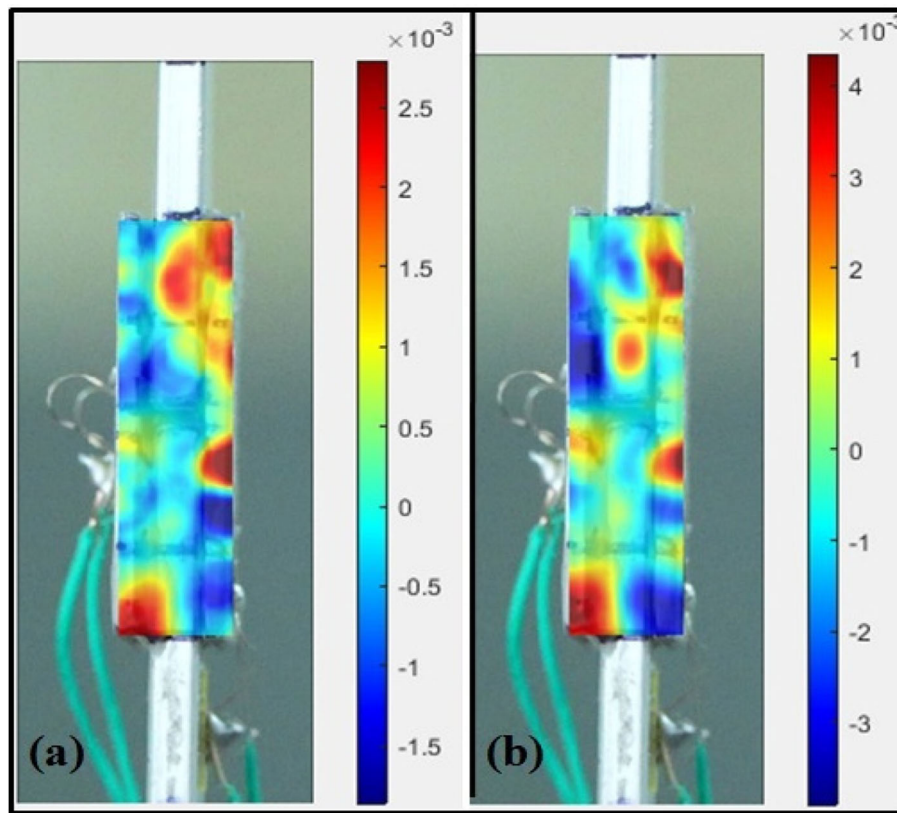


Fig. 19 Shear strain ϵ_{xy} contours along the DSJ for sample S1 for the load of 1.0 kN (elastic load) and 2.45 kN (failure load) using Ncoor software

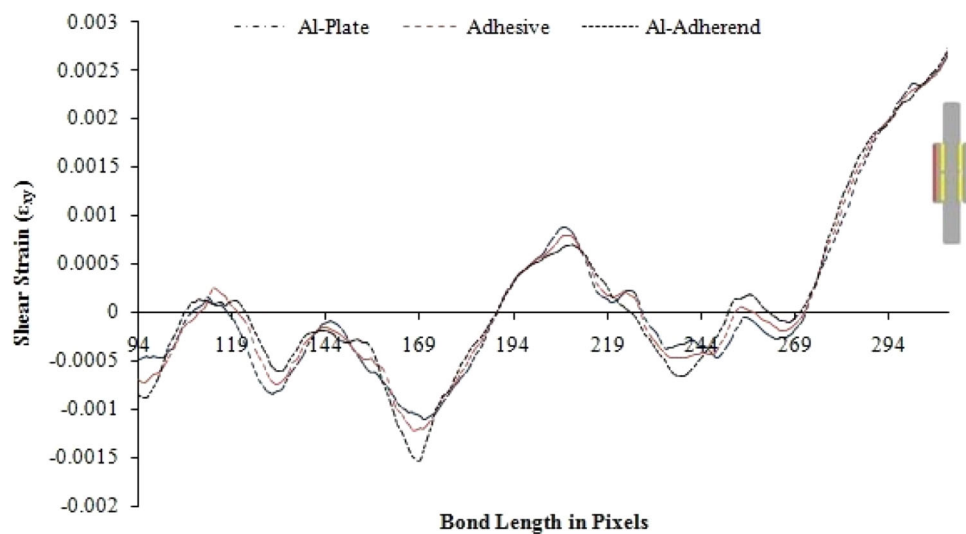


Fig. 20 Variation of shear strain along the left side of bond length in case of plate, adhesive and adherend layer for load of 1.0 kN (elastic load)

suggests that the joint's left side initially resists the applied load through compression, but as the load increases, the joint experiences shear deformation, resulting in tensile shear strain. The right side shows consistent tensile shear strain throughout the bond length, with increasing values toward the bond's end. This indicates that the right side is loaded more directly in tension, reflecting a more straightforward load transfer. From bond length 150-200 pixels, the region exhibits the most

significant increase in shear strain for both sides, particularly in the adhesive and adherend layers. This suggests that the mid-bond region is critical for handling the applied shear load, where most deformation occurs. Figures 22 and 23 show that the plate on the left side is largely under compressive shear strain, whereas the right side experiences significant tensile strain. This asymmetry is expected in bonded joints under load, as one side bears more tension while the other experiences

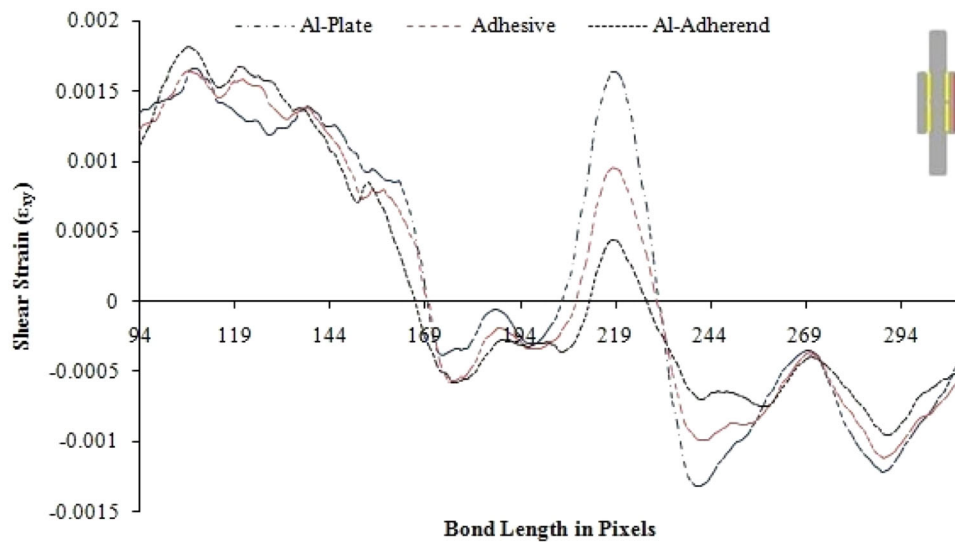


Fig. 21 Variation of shear strain along the right side of bond length in case of plate, adhesive and adherend layer for load of 1.0 kN (elastic load)

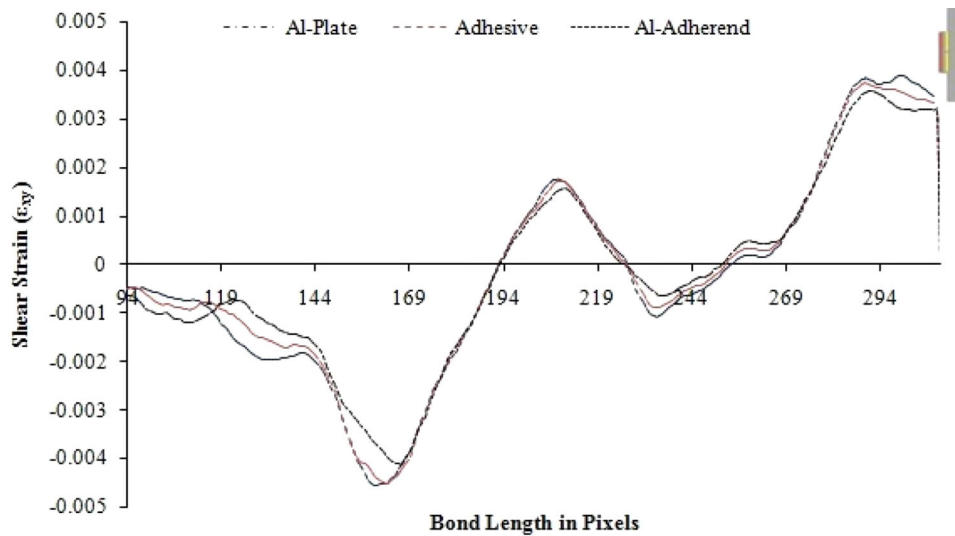


Fig. 22 Variation of shear strain along the left side of bond length in case of plate, adhesive and adherend layer for load of 2.45 kN (failure load)

compression. The sharp rise in tensile strain toward the right-side failure points suggests that this side is stretching significantly under load while the left side is compressed. The adhesive layer bears significant shear strain because it acts as the load transfer medium between the plate and adherend. On the left side, the adhesive is compressed under the applied load, while on the right side, it stretches. The adhesive layer's role in load distribution is critical, and the strain progression shows that the adhesive is subjected to increasing deformation as the joint approach's fails. The adherend layer works in tandem with the adhesive to bear the shear forces. The compressive strain on the left side and tensile strain on the right side reflect the adherend's role in accommodating lateral loads transferred from the adhesive and plate. The strain distribution indicates significant deformation near the failure load, especially on the right side, where the adherend is stretched. The asymmetry in shear strain between the left and right sides indicates the

complex load distribution in double-shear joints. The left side resists the load through compression, while the right side stretches under tension. This behavior is expected as the load is not perfectly symmetric across the joint, leading to different deformation mechanisms on each side. At the middle of the bond length (~150-200 pixels), all layers' shear strain values are highest for both the left and right sides. The adhesive and adherend layers show the most significant strain, suggesting that the middle bond region bears the highest load and is the most critical for joint failure. At the end of the bond length (280-300 pixels), the right side shows peak strain values toward the bond's end, especially in the adhesive and adherend layers, indicating significant tensile shear strain as the material approaches failure. The left side also shows high compressive strain in these regions. The plate exhibits moderate tensile shear strain on the right side, suggesting it stretches but does not deform as much as the adhesive and adherend layers. However,

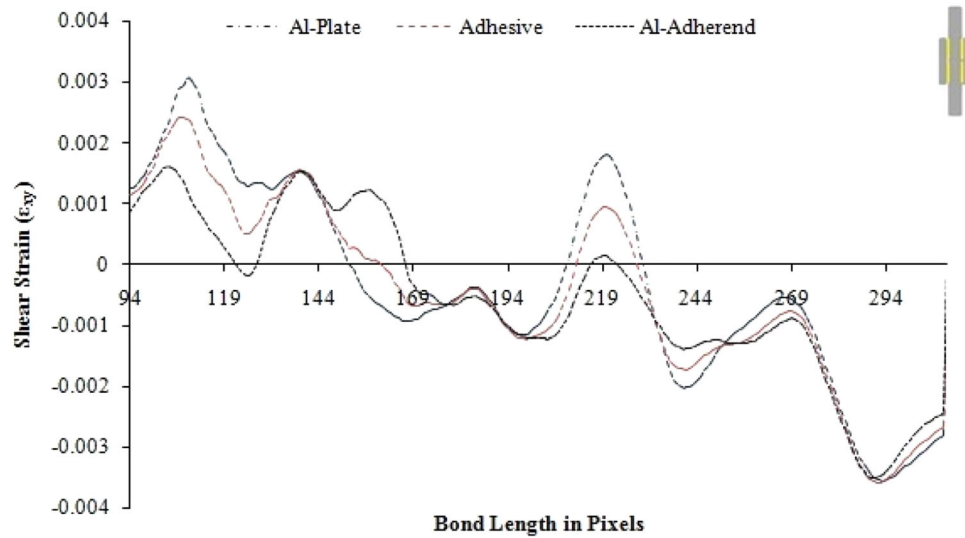


Fig. 23 Variation of shear strain along the right side of bond length in case of plate, adhesive and adherend layer for load of 2.45 kN (failure load)

the compressive strain on the left side indicates that the plate resists significant lateral forces, contributing to the joint's overall failure resistance. The adhesive layer shows the highest shear strain values, particularly on the right side, where it acts as the primary load transfer medium. Its deformation suggests that the adhesive plays a crucial role in determining the joint's failure point. The adherend also shows significant strain, particularly in the middle bond region, indicating that this layer bears a substantial portion of the shear load. The left side of the bond consistently shows compressive shear strain, particularly in the adhesive and plate layers. The compressive strain deepens along the bond length, with the highest values observed near the middle bond region, suggesting that this side is more heavily loaded in compression. The right side experiences increasing tensile shear strain, particularly in the adhesive and adherend layers. The peak strain values are observed near the bond's end, indicating that the right side is most stretched, leading to joint failure at higher loads. Just prior to failure, shear strain (ϵ_{xy}) values along the central axis of the adhesive bond thickness were acquired using digital image correlation (DIC). Figures 19(a) and (b) depict the distributions of shear strain (ϵ_{xy}) along the x-axis and y-axis, respectively, as force increases from the adhesive layer's midpoint within the elastic limit up to the failure load. Figures 20, 21 and 23 illustrate the variations in shear strain (ϵ_{xy}) along the length of the adhesive bond. Upon examining the shear strain distributions and DIC results (ϵ_{xy}), it was noted that the initial portion of the adhesive bond length exhibited a decline in shear strain values. In contrast, the maximum shear strain value was observed in the central region of the adhesive layer.

3.2.4 Normal Strains at Critical Point of Adhesive Layers. Normal strain values at critical locations (a, b, c, d, e, f, g, and h) of adhesive layers under different load conditions are shown in Fig. 8. The normal strain along the y-axis of the adhesive joint across the longitudinal axis is measured using the DIC method. The normal strain values obtained along the plane of shear for loads of 1.0 kN and 2.45 kN are shown in Fig. 24.

From the Fig. 24, it is observed that the normal strain at various critical locations generally tends to increase as the load increases. This behavior aligns with the intuitive understanding

that higher loads cause greater deformation in adhesive layers. At 1.0 kN, the normal strain values range from -0.00222 (point h) to 0.002175 (point b), indicating a combination of compressive and tensile strains within the adhesive layer. Points a and h exhibit negative strain, indicating compression at these critical points, while points such as b, d, and f show positive strain, indicating tension. At a higher load of 2.45 kN, the normal strain values generally increase, with values ranging from -0.00321 (point h) to 0.00325 (point b). Most points exhibit an increase in normal strain, highlighting greater tensile deformation across the adhesive layer as the load approaches the failure point. Similarly, compressive strain becomes more pronounced at points a and h, indicating increased compression under the higher load. At 1 kN load, the strain distribution shows a mix of compression and tension throughout the adhesive layer. The relatively lower values of strain indicate that the adhesive is still within its elastic deformation range and is not yet experiencing significant damage or plastic deformation. At 2.45 kN, close to the failure load, the normal strain values increase significantly at most points, indicating greater deformation of the adhesive layer. Points under tension experience more stretching, while points under compression show deeper compressive strain. The adhesive layer is approaching its strain capacity, and the increased strain at critical points suggests that the material is nearing failure. Point a undergoes a transition from compression at 1.0 kN to slight tension at 2.45 kN. This behavior indicates localized stress redistribution as the load increases, potentially due to material softening or reorientation of stress paths within the adhesive. Point b exhibits the highest tensile strain at both loads. The sharp increase in tensile strain at this point suggests that it may be a critical location for stress concentration and potential crack initiation in the adhesive layer under higher loads. Point h consistently shows the highest compressive strain at both loads, with strain values increasing significantly as the load approaches failure. This point may be experiencing buckling or material compaction, indicating that it is a location of high compressive stress within the adhesive layer.

At 1.0 kN, the adhesive experiences moderate tensile and compressive strains, indicating that the joint is still within its

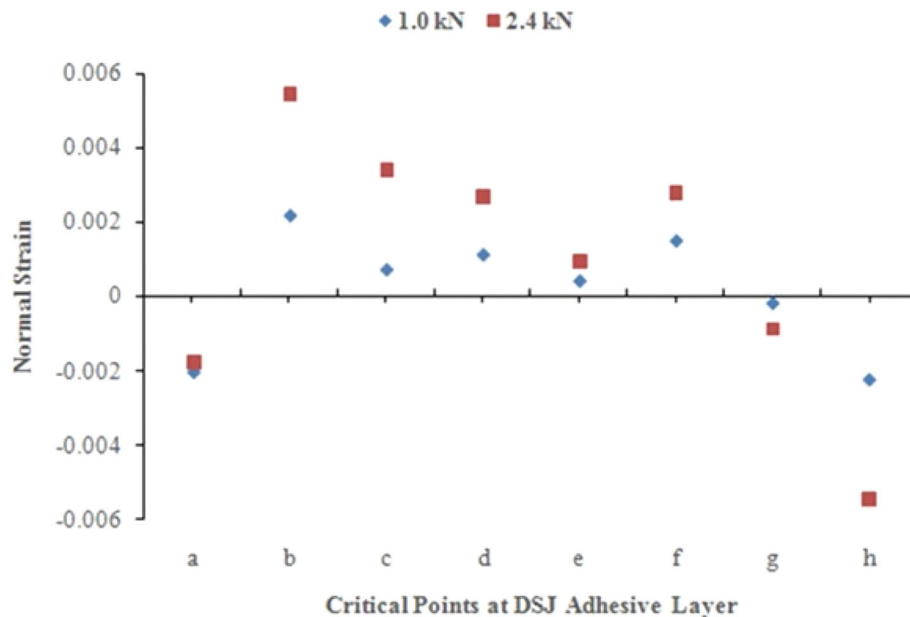


Fig. 24 Variation of normal strain at critical points in adhesive layers

elastic range. At 2.45 kN, the strain values increase significantly, particularly at points b (tensile) and h (compressive), indicating that the adhesive layer is nearing failure. The tension-compression asymmetry highlights the complex stress distribution within the adhesive, with some regions undergoing significant stretching while others are subjected to deep compression.

3.2.5 Shear Strain at the Critical Point of Adhesive Layers. Shear strain values at critical locations of adhesive layers under different load conditions. The shear strain along the y-axis of the adhesive joint, i.e., across the longitudinal axis, is measured using DIC method. The strain values obtained along the plane of shear for loads of 1.0 and 2.45 kN are shown in Fig. 25.

Figure 25 shows that the Shear strains at critical locations generally increase with increasing load. This corresponds to the expected behavior where higher loads lead to increased deformation due to shearing forces acting on the plate. The shear strain values at 1.0 kN range from -0.0007 at point a to 0.002721 at point h. A combination of positive and negative shear strain indicates the direction and magnitude of shear deformation. Some points like b, e, and h show positive shear strain, indicating shearing in one direction, while points like a, c, f, and g show negative shear strain, indicating shearing in the opposite direction. At a higher load of 2.45 kN, the range of shear strain values broadens from -0.00265 at point g to 0.003325 at point h. The absolute values of strain increase for most points, showing that shear deformation intensifies as the load increases. At 1.0 kN, the adhesive experiences moderate shear deformation. There is a mix of positive and negative shear strain, indicating that different regions of the adhesive are experiencing different types of shear deformation (tension vs. compression). The small strain values suggest the adhesive layer is within its elastic range, showing minimal signs of damage. At 2.45 kN, close to the failure load, the shear strain values increase for most points, especially those already under tension (like points e and h). The increased shear strain signifies the adhesive is undergoing greater shear deformation, poten-

tially approaching its strain capacity. Points experiencing compressive shear strain (like g and f) show a significant increase in compression, indicating stress concentration and possible material failure in those areas under higher load. Point h consistently exhibits the highest positive shear strain at both loads. The significant increase from 0.002721 to 0.003325 at higher load suggests this point experiences concentrated shear tension, potentially leading to material failure due to excessive shear deformation. Point g shows a substantial increase in compressive shear strain from -0.00054 to -0.00265 , indicating a high compression zone in the adhesive layer. This behavior suggests that localized stress could lead to adhesive failure in compression under high load. Point d shifts from slight tension (0.000516) at 1.0 kN to compression (-0.00046) at 2.45 kN. This reversal suggests stress redistribution under a higher load, possibly indicating that other regions are bearing more shear tension while this point experiences compression. At 1.0 kN, the adhesive experiences moderate shear deformation, with a mix of shear tension and compression distributed across the adhesive. At 2.45 kN, the shear strains increase, particularly at points like h (tensile strain) and g (compressive strain), indicating areas of high stress and potential failure as the load approaches its maximum. The direction of shear strain (positive vs. negative) provides critical information on how the adhesive material is responding to the load, with certain regions undergoing significant shear tension and others undergoing shear compression.

Upon comparing the normal and shear strain results with existing literature, we observe that the strain magnitudes in the adhesive layers and plates align well with those reported by (Ref 48). However, our results show asymmetrical strains in the elastic phase, particularly at locations b and f. This asymmetry is corroborated by DIC analysis, which reveals localized strain concentrations likely due to geometric imperfections and material inhomogeneities. Similar findings are discussed in (Ref 49), where boundary condition inconsistencies and microstructural effects were identified as contributing factors.

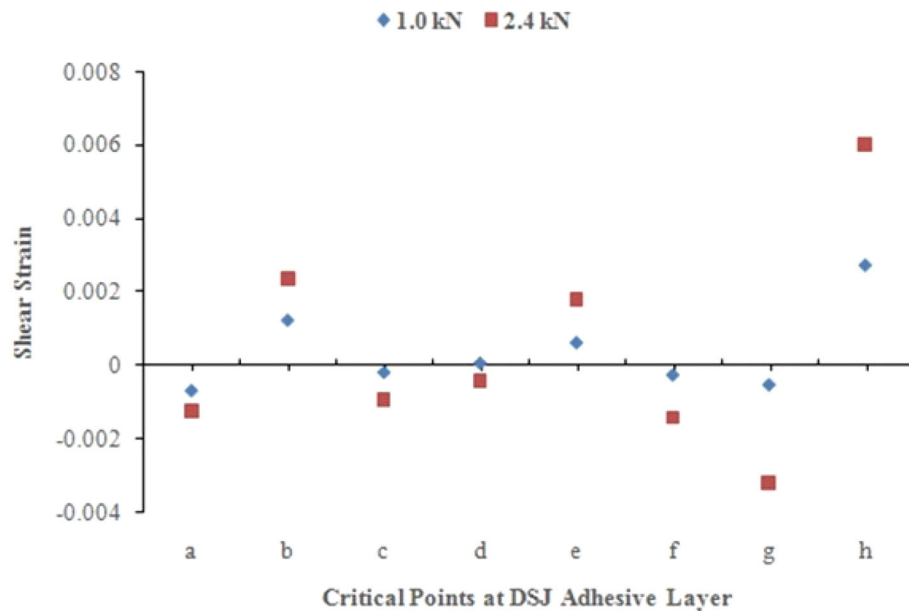


Fig. 25 Variation of shear strain at critical points in adhesive layers

Earlier studies have recognized the importance of the adhesive layer in strain gauge mounting but often focused on limited aspects, such as specific types of adhesives or loading conditions. Our study expands on this by comprehensively analyzing the adhesive layer's role using experimental methods, FEA simulations, and analytical models. This study explores a broader range of adhesive properties and loading scenarios, providing a more holistic understanding. Many previous studies relied predominantly on experimental approaches or FEA simulations without comprehensively integrating both methodologies. The present study integrates experimental data with FEA simulations and analytical models, allowing for a more robust validation and cross-verification of results. This multi-method approach offers a more detailed and accurate depiction of strain transfer mechanisms. DIC technology has been underutilized in earlier studies focusing on strain gauge applications, with more reliance on traditional strain measurement techniques. By incorporating DIC, we achieve high-resolution, non-contact strain measurements that enhance the accuracy and spatial resolution of the data, providing new insights into strain distribution and transfer.

3.3 Finite Element Analysis Results

A nonlinear finite element module was considered for performing finite element analysis using ANSYS software on double strap joint samples. Double strap joint samples are subjected to tensile loads. The results obtained from finite element analysis, such as displacement, normal strain along the axis, shear strain along a plane, maximum principal strain, and maximum shear strain, are shown in Figs. 26, 27, 28, 29, 30, 31, 32 and 33.

3.3.1 Finite Element Results Validation. Load versus deflection curves obtained from the experiment and finite element analysis are depicted in Fig. 34, respectively. The case is a double strap joint model subjected to tensile loading.

Figure 34 illustrates that all samples (DSJ S1, DSJ S2, DSJ S3) exhibit a linear response initially, where the displacement increases slowly with small loads. After this initial region, there

is a noticeable linear trend between load and displacement as the applied load increases. This linearity suggests elastic deformation in the joint structures during this phase. DSJ S1 reaches its peak load at around 1.3 kN, with corresponding experimental displacement of 0.112 mm and FEA displacement of 0.103 mm. The sample then experiences a load drop, indicating potential yielding or failure. DSJ S1 shows good agreement between experimental and FEA data throughout the load range. The differences in displacement values are minimal, indicating that the FEA model is well-calibrated for this sample. DSJ S2 reaches its peak load at approximately 1.74 kN, showing greater load-bearing capacity than DSJ S1. The corresponding displacements are 0.124 mm (experiment) and 0.115 mm (FEA). DSJ S2 has larger discrepancies between experimental and FEA results, particularly at higher loads. The FEA results predict slightly smaller displacements than the experimental ones. DSJ S3 has the highest load capacity, peaking at 2.45 kN, with displacements of 0.152 mm (experiment) and 0.144 mm (FEA). This reflects DSJ S3's strength and its cohesive mode of failure. DSJ S3, like DSJ S1, shows relatively good agreement, but the FEA predictions are consistently lower than the experimental values. This may indicate that the joint has a potential slip at the interface during the experiments not considered in the simulation.

The finite element model may assume simplified material properties, leading to discrepancies in displacement predictions, especially at higher loads where nonlinear material behavior, such as localized damage, might occur in the actual experiments. In real experiments, joints might experience some slip or uneven load distribution, especially at the interface between components. FEA models often assume idealized contact, which could explain the smaller displacement predictions compared to experimental results. Variations in the geometric tolerances of the joint samples during specimen preparation can affect the experimental results. These tolerances might not be accounted for in the FEA model, leading to slight deviations in the load-displacement response. The method of load application in the experiments might introduce additional factors, such as

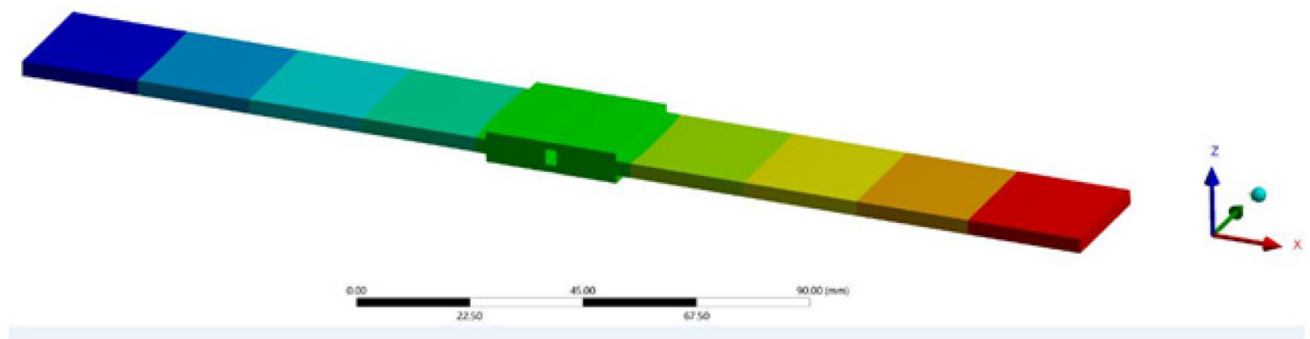


Fig. 26 Contour representing variation of displacement

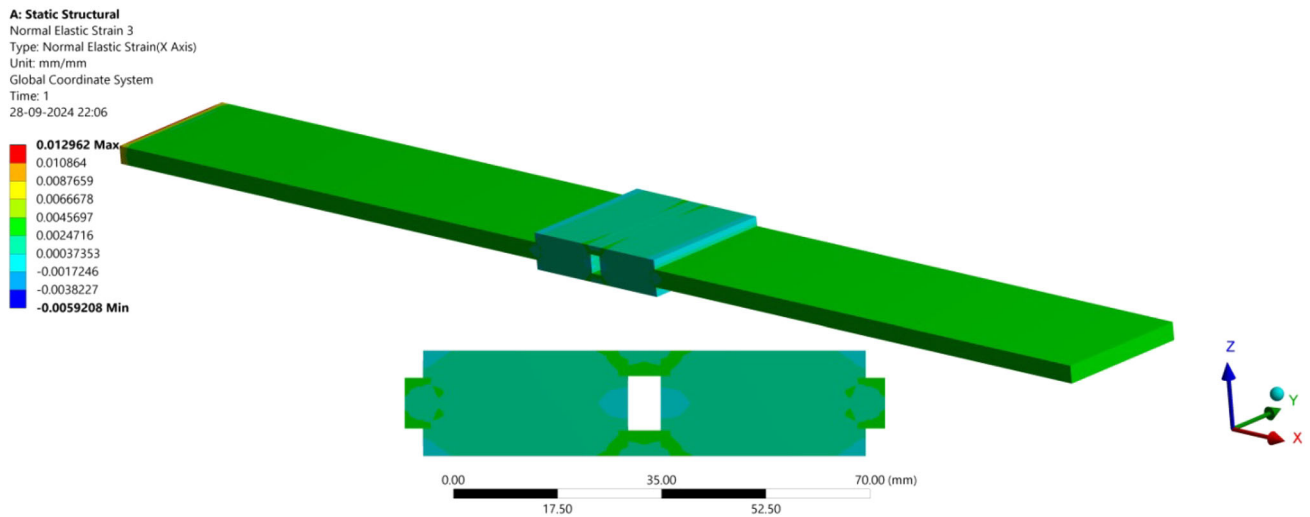


Fig. 27 Variation of normal strain in X-direction

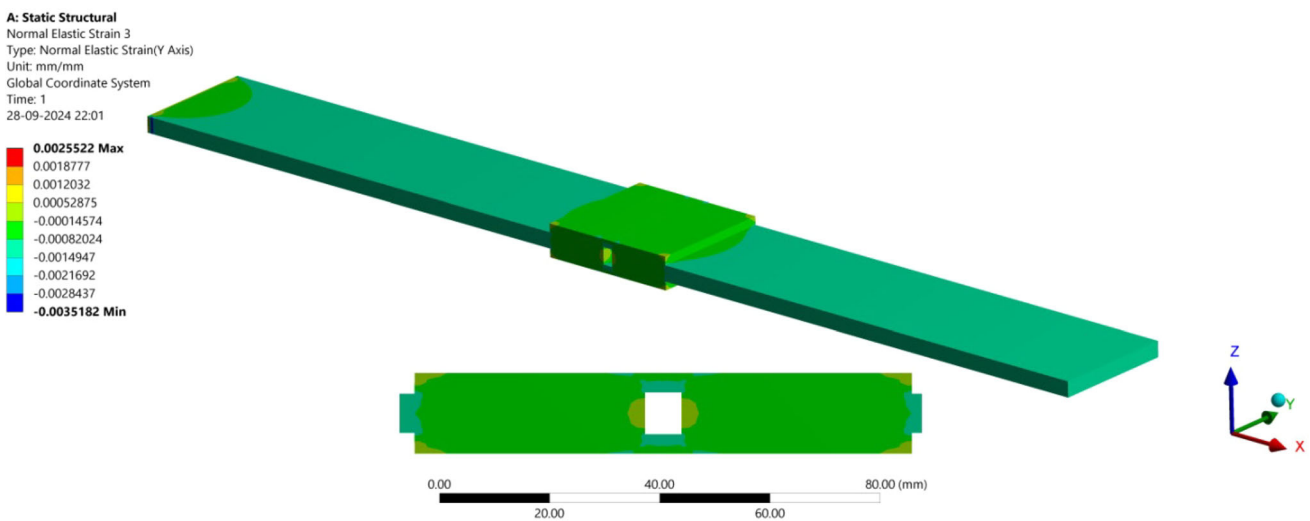


Fig. 28 Variation of normal strain in Y-direction

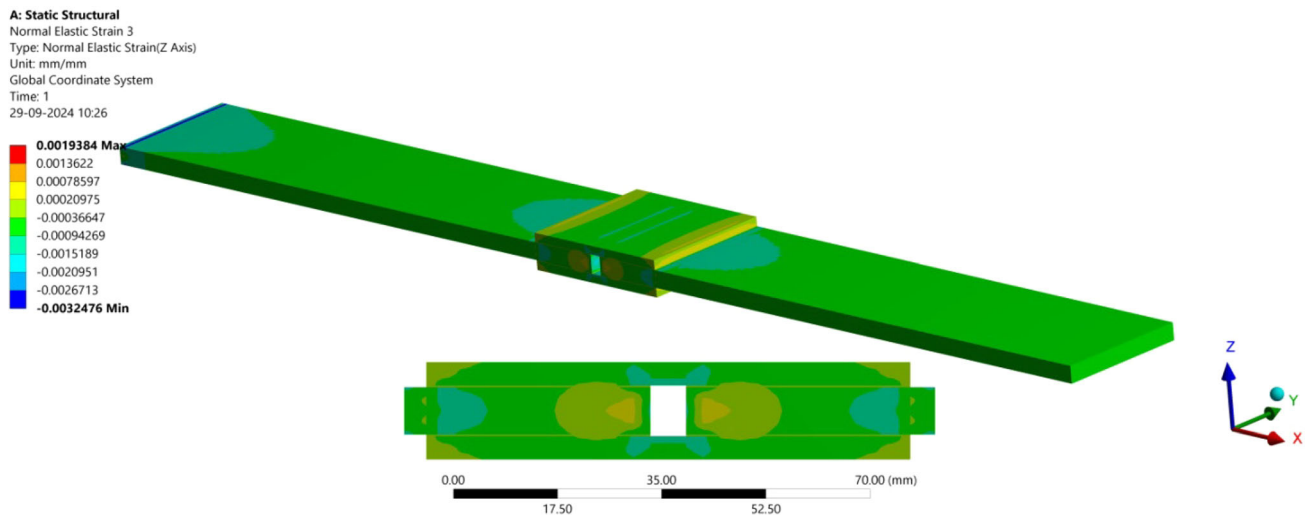


Fig. 29 Variation of normal strain in Z-direction plane

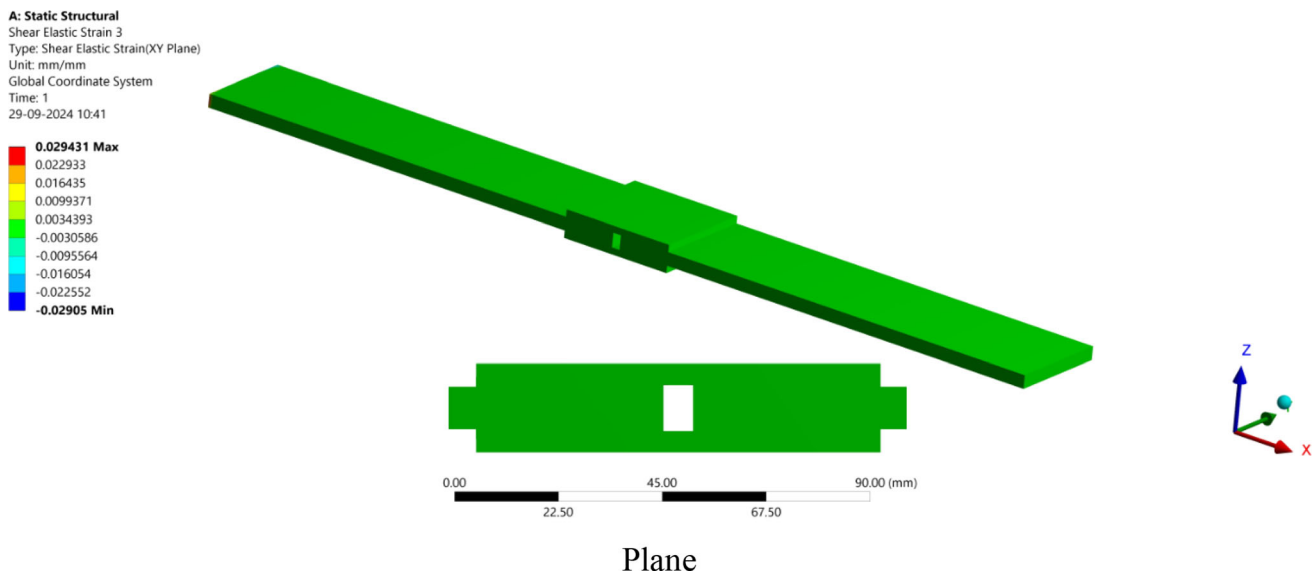


Fig. 30 Variation of shear strain in X-Y

misalignment or off-axis loading. In FEA, the load is applied in a perfectly controlled manner, which may lead to smaller displacements for the same load levels.

3.3.2 Comparison of FEM Model and DIC Result for Normal Strain Values. The comparison of normal strain values at critical locations of the adhesive layer, as shown in Table 4, and FEM variations compared within the elastic limit is shown in Fig. 35.

There is a reasonable correlation between the DIC and ANSYS results. However, significant variations are observed at certain locations, particularly at locations b, d, e, and f. Negative values are observed at the a, g, and h points. These variations could be attributed to:

- **Modeling Assumptions:** Simplifications or assumptions made in the ANSYS model that do not accurately capture the actual behavior of the adhesive layer.

- **Stress Concentrations:** Localized areas of high stress or strain that may not be captured accurately in the simulation.
- **Additional Loading Effects:** The ANSYS simulation currently considers only tensile loading, neglecting other combined effects such as compression and bending. This can lead to the appearance of negative values in the results.

Overall, the ANSYS simulation provides a good approximation of the normal strain values, but certain improvements in the modeling approach could enhance the accuracy and correlation with experimental results.

3.4 DIC Results Validation with Analytical Models

Figure 36 illustrates a comparison of normalized adhesive shear strain distribution values obtained from analytical models

A: Static Structural
Maximum Shear Elastic Strain
Type: Maximum Shear Elastic Strain
Unit: mm/mm
Time: 1
29-09-2024 10:59

0.032139 Max
0.028578
0.025018
0.021457
0.017896
0.014335
0.010774
0.0072129
0.003652
9.108e-5 Min

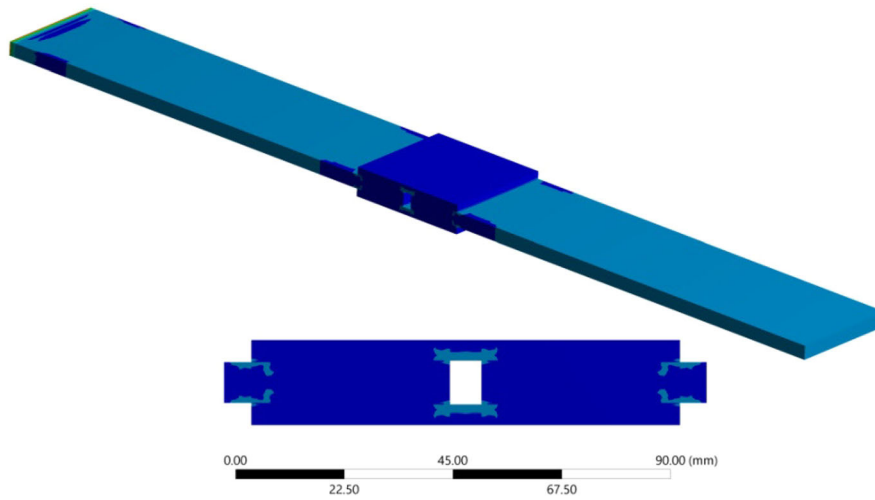


Fig. 31 Variation of maximum shear elastic strain

A: Static Structural
Maximum Principal Elastic Strain
Type: Maximum Principal Elastic Strain
Unit: mm/mm
Time: 1
29-09-2024 11:01

0.02192 Max
0.019487
0.017054
0.014621
0.012188
0.0097555
0.0073225
0.0048894
0.0024564
2.3427e-5 Min

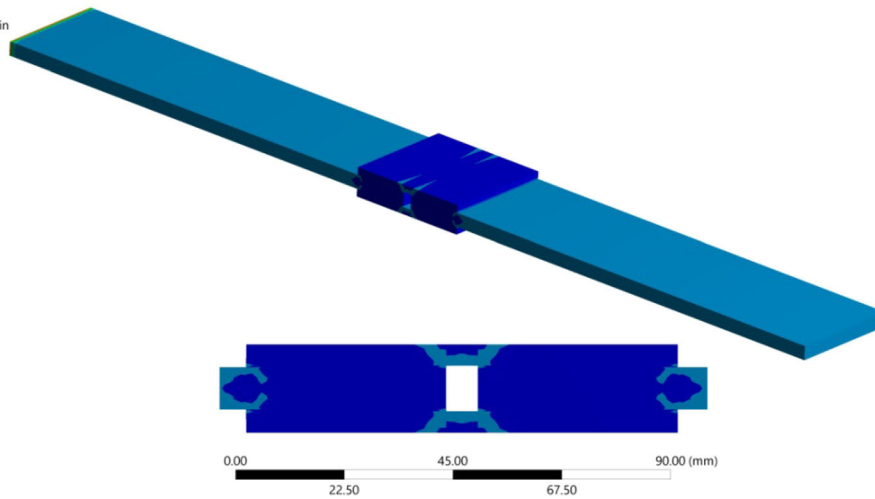


Fig. 32 Variation of maximum principal elastic strain

A: Static Structural
Equivalent Elastic Strain
Type: Equivalent Elastic Strain
Unit: mm/mm
Time: 1
29-09-2024 11:01

0.02142 Max
0.019056
0.016692
0.014328
0.011964
0.0095996
0.0072355
0.0048715
0.0025074
0.00014337 Min

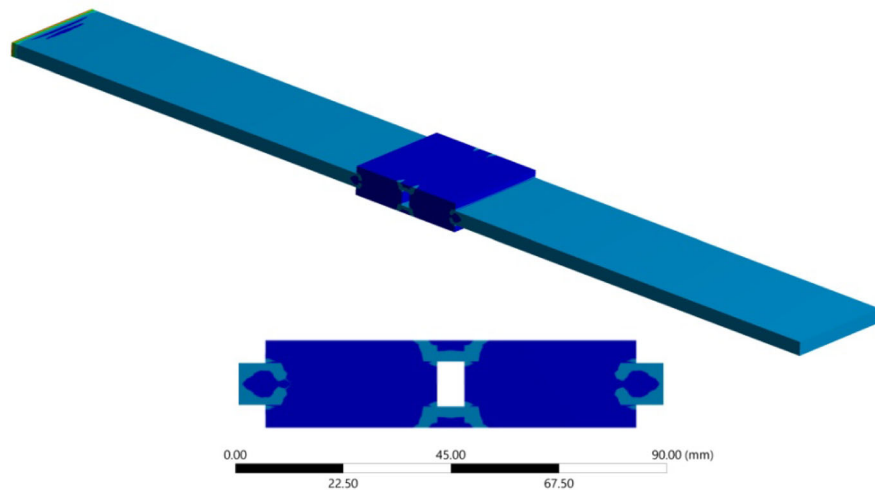


Fig. 33 Variation of von Mises strain

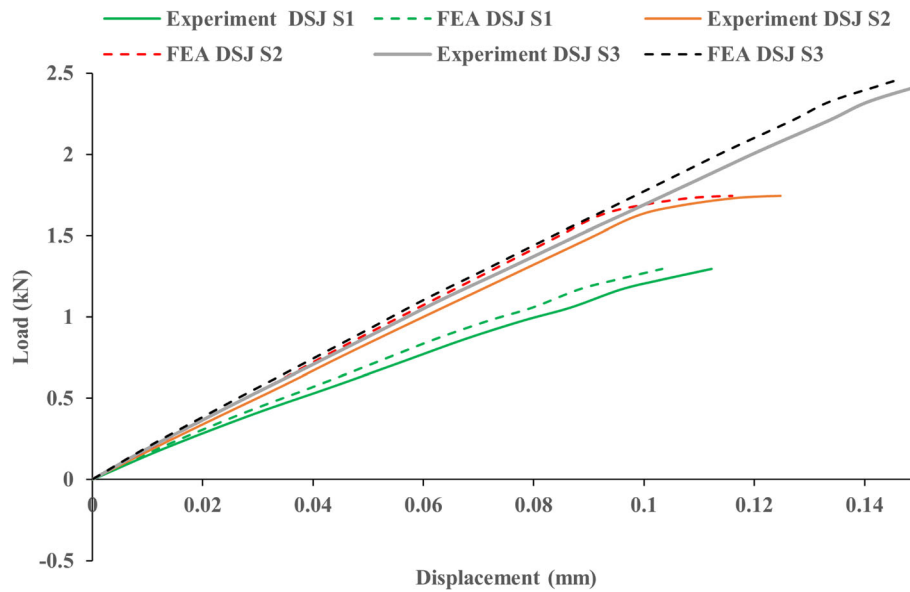


Fig. 34 Comparison of experimental and FEA results showing load versus deflection curve of DSJ S1, DSJ S2, and DSJ S3 specimen

Table 4 Comparison of FEM model and DIC result for normal strain values:

Normal strain (mm/mm)	Strain obtained at different locations of adhesive layer							
	a	b	c	d	e	f	g	h
DIC results	−0.00202	0.002175	0.00722	0.001121	0.00419	0.001496	−0.00018	−0.00222
ANSYS results	0.002023	0.002111	0.002244	0.002308	0.00235	0.00223	0.002065	0.002024

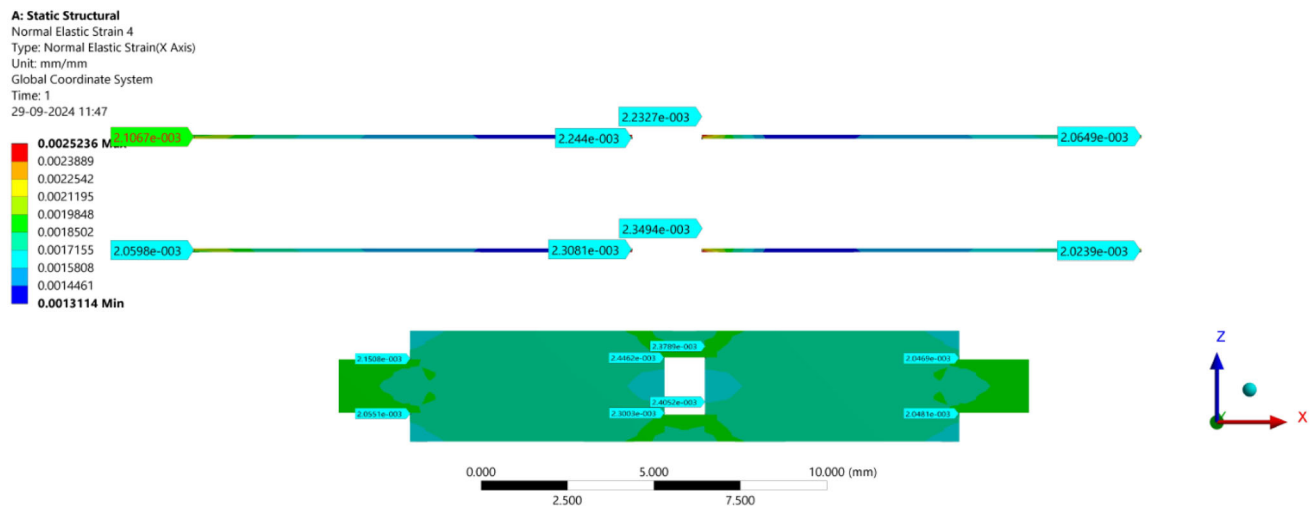


Fig. 35 Finite element analysis results obtained at critical locations of adhesive layer.

(Volkersen and Tsai₂) with experimental results from DIC (digital image correlation) where x is points considered along overlap length, c is overlap length, γ_a is shear strain of adhesive and γ_{avg} is average shear strain of adhesive.

All three approaches display a nonlinear trend, with a noticeable increase in the normalized shear strain (γ_a/γ_{avg}) near the edges of the overlap, which is typical in adhesive joints due to stress concentrations. Experimental shear strain distribution is generally smoother, reflecting real adhesive joint behavior

where material imperfections, load distribution, and adhesive ductility influence the strain profile. By comparing the analytical model predictions (Volkersen and Tsai₂) with the experimental DIC results, we can assess the accuracy and reliability of the analytical models in predicting adhesive shear strain. A close agreement between the analytical predictions and DIC experimental results indicates the models' effectiveness in capturing the adhesive behavior. Deviations highlight areas where the analytical models might need refinement or

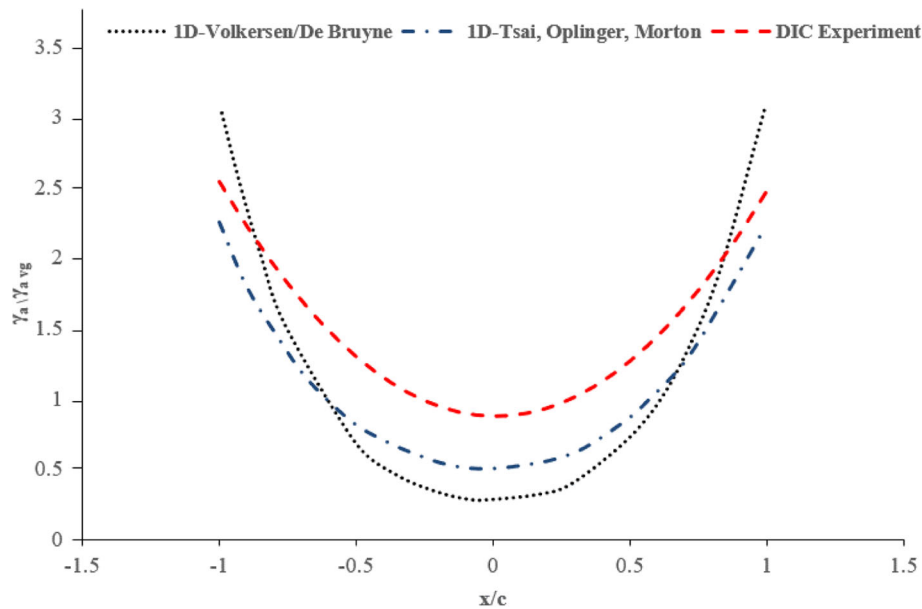


Fig. 36 Comparison of normalized adhesive shear strain distribution values

where experimental conditions may differ from the assumed model conditions.

3.4.1 Coherent Points. The Volkersen and Tsai models provided reasonable predictions for overall strain transfer trends and were in good agreement with experimental data for moderate loading conditions. These models effectively captured the primary factors affecting strain transfer, such as adhesive stiffness and thickness.

3.4.2 Incoherent Points. Under high-loading conditions, the analytical models tended to underestimate the strain values compared to the experimental results. This discrepancy is likely due to the simplifications inherent in analytical models, which may not fully account for complex stress distributions and potential nonlinearities in the adhesive behavior.

The assumption of perfect bonding and uniform material properties might not fully capture the complexities of actual adhesive layers, leading to overestimations in localized strain regions. Simplified assumptions regarding adhesive behavior and stress distribution may result in less accurate predictions under extreme loading conditions. Variations in adhesive material properties compared to the characterized values used in simulations and analytical models can introduce discrepancies. Factors such as adhesive curing, environmental conditions, and potential defects in the adhesive layer can impact performance. Experimental measurements using DIC, while highly accurate, can still be subject to uncertainties due to calibration errors, lighting conditions, and surface preparation of the test samples. These factors might contribute to differences between experimental and numerical results.

4. Conclusions

The present research study gives valuable insights into the behavior of double strap joint specimens (DSJ S1, DSJ S2, and DSJ S3) with varying adhesive layer thicknesses under tensile loads. The innovative aspect of this research lies in exploring

how adhesive layer thickness influences the bond strength and strain transfer of such joints. The research investigates the failure mechanisms of DSJ specimens, uncovering multiple stages in the process, from initial deformation to microcrack development and eventual catastrophic failure. This comprehensive examination of failure modes and their progression offers a nuanced perspective on the behavior of these joints, which can be pivotal in developing strategies for failure prevention and enhanced structural resilience. The analysis of shear and normal strains within the adhesive layers reveals intricate strain patterns and variations across critical locations. Finite element analysis has been performed to validate the experimental results. Several conclusions can be drawn regarding the behavior of the adhesive layers and double strap joint specimens.

- (1) The bond strength of double strap joint specimens varies with adhesive layer thickness, with DSJ S3 (thinner adhesive layer) having the highest load capacity and DSJ S1 (thick adhesive layer) showing lower strength than DSJ S2, highlighting the critical role of adhesive thickness.
- (2) The load-displacement curves of all three specimens exhibit a linear elastic region followed by a sudden failure, without significant plastic deformation, ultimately leading to brittle fracture.
- (3) DSJ S3 stands out with the highest load-carrying capacity and a prolonged deformation, likely due to even adhesive layer distribution showing cohesive failure.
- (4) The failure mechanism in double strap joint specimens involves stages such as initial adherend deformation, microcrack development, material softening, and eventual catastrophic failure, with local and progressive failures causing intermittent load decreases during mode transitions.
- (5) As the bond length increases, the plate layer's displacement shows a trend, indicating more uniform deformation.

tion along its length. Due to the adhesive's elastoplastic behavior, the middle region of the lap length exhibits higher displacements than the ends.

- (6) The analysis shows uneven shear strain distribution along the adhesive joint, peaking in the middle and decreasing toward the unloaded end, indicating non-uniform shear stress, with greater stress and deformation at the loaded end.
- (7) The adhesive layer is vital for transmitting shear forces between the plate and adherend, as evidenced by increasing shear strain toward the right side, signifying effective force transfer and deformation within the adhesive layer, essential for preventing shear failures along the bond length.
- (8) The joint's layers display distinct shear strain responses, with the plate having lower shear strain due to its stiffness, while the adhesive and adherend layers exhibit higher shear strain due to their differing mechanical properties and bond strengths.
- (9) The smooth decrease in shear strain from left to right indicates layer compatibility, minimizing localized stress concentrations and enhancing joint stability, which is a desirable characteristic for preventing joint failure.
- (10) Normal strain analysis reveals symmetric strains in the adhesive and adherend layers around the bond's midpoint, suggesting balanced structural behavior and load distribution, while the plate consistently experiences compression along its length.
- (11) Shear and normal strains increase with increasing load values, which is expected as higher loads induce greater deformation in the joint. These strains can transition from compression to tension as the load approaches the failure load.
- (12) Adhesive layers exhibit increasing normal strain with load, varying at critical locations due to stress distribution and material differences, occasionally showing non-monotonic behavior, indicating localized effects like stress concentrations. Adhesive layers display linear elastic behavior until specific load limits before potential plastic deformation or failure mechanisms may occur.
- (13) Shear strain increases with load as expected, varies across critical locations due to non-uniform force distribution and geometric factors, and experiences significant changes at specific load levels, potentially due to stress concentrations or shifts in load transfer mechanisms. Negative shear strains and nonLinear behavior indicate complex interactions between forces, material properties, and geometry within the adhesive layers.
- (14) Transition regions in strain patterns signify deformation changes, potentially related to material boundaries or load distribution variations, and may reveal anomalies or specimen irregularities.
- (15) The agreement between experimental and finite element analysis (FEA) data is generally good for all DSJ specimens, suggesting that the FEA model used is a reasonable representation of the physical behavior.
- (16) The thickness of adhesive layer should be less due to as the layer thickness increases the inner layers are not cured properly results to elastic plastic strain.

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References

1. F. Delzendehrooy, A. Akhavan-Safar, A.Q. Barbosa, R. Beygi, D. Cardoso, R.J.C. Carbas, E.A.S. Marques, and L.F.M. Da Silva, A comprehensive review on structural joining techniques in the marine industry, *Compos. Struct.*, 2022, **289**, p 115490. <https://doi.org/10.1016/j.compstruct.2022.115490>
2. S. Lathabai, 20 - Joining of aluminium and its alloys, Editor(s): Roger Lumley, In Woodhead Publishing Series in Metals and Surface Engineering. Fundamentals of Aluminium Metallurgy, Woodhead Publishing, 2011, Pages 607-654, <https://doi.org/10.1533/9780857090256.3.607>
3. M.Z. Sadeghi, J. Weiland, A. Preisler, J. Zimmermann, A. Schiebahn, U. Reissen, and K.U. Schroeder, Damage detection in adhesively bonded single lap joints by using backface strain: Proposing a new position for backface strain gauges, *Int. J. Adhes. Adhes.*, 2020, **97**, p 102494. <https://doi.org/10.1016/j.ijadhadh.2019.102494>
4. T.S. Sahana, V. Kaliveeran, and S. Kundapura, Mechanical characterization of adhesive layer using double strap joint specimens, *Mater. Today: Proc.*, 2022, **66**, p 509–513.
5. F. Findik, N. Mrad, and A. Johnston, Strain monitoring in composite patched structures, *Compos. Struct.*, 2000, **49**(3), p 331–338.
6. A.D. Crocombe, C.Y. Ong, C.M. Chan, M.A. Wahab, and I.A. Ashcroft, Investigating fatigue damage evolution in adhesively bonded structures using backface strain measurement, *J. Adhes.*, 2002, **78**(9), p 745–776.
7. A.G. Solana, A.D. Crocombe, M.A. Wahab, and I.A. Ashcroft, Fatigue initiation in adhesively-bonded single-lap joints, *J. Adhes. Sci. Technol.*, 2007, **21**(14), p 1343–1357.
8. Adams, R. D. Failure strength tests and their limitations, American Society of Metals (ASM) Adhesives Handbook, ASM International (1987)
9. P.D. Ruiz, F. Jumbo, J.M. Huntley, I.A. Ashcroft, and G.M. Swallowe, Experimental and numerical investigation of strain distributions within the adhesive layer in bonded joints, *Strain*, 2011, **47**(1), p 88–104.
10. A. Kaya, M. Tekelioglu, and F. Findik, Effects of various parameters on dynamic characteristics in adhesively bonded joints, *Mater. Lett.*, 2004, **58**, p 3451–3456.
11. R. Jairaja and G.N. Naik, Single and dual adhesive bond strength analysis of single lap joint between dissimilar adherends, *Int. J. Adhes. Adhes.*, 2019, **92**, p 142–153.
12. M. Caliskan, U. Soy, and F. Findik, An analytic investigation to compare the bonded and bolted repairing methods using composite single lap joint, *Sci. Eng. Compos. Mater.*, 2003, **11**(1), p 69–77.
13. F.L. Matthews, P.F. Kilty, and E.W. Godwin, A review of the strength of joints in fibre-reinforced plastics. Part 2. Adhesively bonded joints, *Composites*, 1982, **13**(1), p 29–37.
14. G. Jeevi, S.K. Nayak, and M. Abdul Kader, Review on adhesive joints and their application in hybrid composite structures, *J. Adhes. Sci. Technol.*, 2019, **33**(14), p 1497–1520.
15. P.A. Gustafson, A. Bizard, and A.M. Waas, Dimensionless parameters in symmetric double lap joints: an orthotropic solution for thermomechanical loading, *Int. J. Solids Struct.*, 2007, **44**, p 5774–5795. <https://doi.org/10.1016/j.ijsolstr.2007.01.025>
16. B.A. Bednarczyk, J. Zhang, C.S. Collier, Y. Bansal, and M.-J. Pindera, Analysis tools for adhesively bonded composite joints, part 1: higher-order theory, *Aiaa J.*, 2006, **44**, p 171–180. <https://doi.org/10.2514/1.12061>
17. I. Pires, L. Quintino, J.F. Durodola, and A. Beevers, Performance of Bi-adhesive bonded aluminium lap joints, *Int. J. Adhes. Adhes.*, 2003, **23**, p 215–223. [https://doi.org/10.1016/S0143-7496\(03\)00024-1](https://doi.org/10.1016/S0143-7496(03)00024-1)
18. F.-R. Kong, M. You, and X.-L. Zheng, Three-dimensional finite element analysis of the stress distribution in Bi-adhesive bonded joints, *J. Adhes.*, 2008, **84**, p 105–124. <https://doi.org/10.1080/00218460801952791>

19. M. Shishesaz and M. Hosseini, Effects of joint geometry and material on stress distribution, strength and failure of bonded composite joints: an overview, *J. Adhes.*, 2018, **96**(12), p 1053–1121. <https://doi.org/10.1080/00218464.2018.1554483>
20. Hart-Smith, L. J. (1973). Adhesive-bonded double-lap joints (No. NASA-CR-112235)
21. M.Y. Tsai and J. Morton, An evaluation of analytical and numerical solutions to the single-lap joint, *Int. J. Solids Struct.*, 1994 [https://doi.org/10.1016/0020-7683\(94\)90036-1](https://doi.org/10.1016/0020-7683(94)90036-1)
22. Oplinger, D. W. A layered beam theory for single lap joints. US Army Materials Technology Laboratory (1991)
23. G.P. Zou, K. Shahin, and F. Taheri, An analytical solution for the analysis of symmetric composite adhesively bonded joints, *Compos. Struct.*, 2004 <https://doi.org/10.1016/j.compstruct.2004.01.007>
24. M. Azeem, M. Irfan, M. Masud, G.U. Rehman, H. Ali, M.U. Ali, and J. Kratochvil, Experimental and numerical investigation of effect of static and fatigue loading on behavior of different double strap adhesive joint configurations in fiber metal laminates, *Materials*, 2022, **15**(5), p 1840. <https://doi.org/10.3390/ma15051840>
25. A. Pramanik, A.K. Basak, Y. Dong, P.K. Sarker, M.S. Uddin, G. Littlefair, and S. Chattopadhyaya, Joining of Carbon Fibre Reinforced Polymer (CFRP) Composites and Aluminium Alloys—A Review, *Compos. A Appl. Sci. Manuf.*, 2017, **101**, p 1–29.
26. S.K. Panigrahi and S.V. Nimje, *Design and Analysis of Functionally Graded Adhesively Bonded Joints of FRP Composites*, CRC Press, 2022
27. Y. Zhao, H. Wang, H. Li, J. He, Y. Zhou, J. Liu, and M. Zhu, Competitive Fretting Fatigue Failure Analysis of Al Alloy–CFRP Riveted and Adhesive–Riveted Single-Shear Lap Joints, *Eng. Fail. Anal.*, 2023, **153**, 107538
28. M.A. Dybizbański and K. Rzeszut, Experimental and Theoretical Investigation of Galvanized Steel and Fiber-Reinforced Polymer Composites Textile Adhesive Double Lap Joints, *Adv. Sci. Technol. Res. J.*, 2023, **17**(4), p 110–120.
29. H. Chen, J. Na, D. Wang, D. Kong, and X. Zhang, Numerical Simulation and Failure Experiment of Hygrothermal Aged CFRP Single and Double Lap Joints, *Thin-Walled Struct.*, 2023, **188**, 110786
30. X. Wei, H.S. Shen, and H. Wang, Fracture Failure Prediction for Composite Adhesively Bonded Double Lap Joints by an Experiment-Based Approach, *Int. J. Adhes. Adhes.*, 2022, **114**, 103110
31. J. Rocha, J. Sena-Cruz, and E. Pereira, Tensile Behaviour of CFRP-Glass Adhesively Bonded Connections: Double-Lap Joint Tests and Numerical Modelling, *Eng. Struct.*, 2022, **260**, 114212
32. H. Özer and Ö. Öz, Three-Dimensional Finite Element Analysis of Bi-Adhesively Bonded Double Lap Joint, *Int. J. Adhes. Adhes.*, 2012, **37**, p 50–55.
33. R. Iyer Kumar, P. Jaiswal, and W. De Waele, Fatigue Damage and Life Evaluation of Thick Bi-Material Double Strap Joints for use in Marine Applications, *Fatigue Fract. Eng. Mater. Struct.*, 2022, **45**(7), p 2099–2111. <https://doi.org/10.1111/ffe.137010.1111/ffe.13709>
34. H.R. Majidi, S.M.J. Razavi, and F. Berto, Failure Assessment of Steel/CFRP Double Strap Joints, *Metals*, 2017, **7**(7), p 255. <https://doi.org/10.3390/met7070255>
35. M.Y. Tsai and J. Morton, An Investigation into the Stresses in Double-Lap Adhesive Joints with Laminated Composite Adherends, *Int. J. Solids Struct.*, 2010, **47**(24), p 3317–3325.
36. A. Li, H. Wang, H. Li, D. Kong, and S. Xu, Estimation of Bond Strength and Effective Bond Length for the Double Strap Joint Between Carbon Fiber Reinforced Polymer (CFRP) Plate and Corroded Steel Plate, *Polymers*, 2022, **14**(15), p 3069. <https://doi.org/10.3390/polym14153069>
37. Y. Wei, X. Jin, Q. Luo, Q. Li, and G. Sun, Adhesively Bonded Joints—A Review on Design, Manufacturing, Experiments, Modeling and Challenges, *Compos. Part B: Eng.*, 2024 <https://doi.org/10.1016/j.compositesb.2024.111225>
38. K. Leffler, K.S. Alfredsson, and U. Stigh, Shear Behaviour of Adhesive Layers, *Int. J. Solids Struct.*, 2007 <https://doi.org/10.1016/j.ijsolstr.2006.04.036>
39. S.W. Khoo, S. Karuppanan, and C.S. Tan, A Review of Surface Deformation and Strain Measurement Using Two-Dimensional Digital Image Correlation, *Metrol. Meas. Syst.*, 2016, **23**(3), p 461–480.
40. S. Yoneyama, Basic Principle of Digital Image Correlation for In-Plane Displacement and Strain Measurement, *Adv. Compos. Mater.*, 2016, **25**(2), p 105–123.
41. A.J. Comer, K.B. Katnam, W.F. Stanley, and T.M. Young, Characterising the Behaviour of Composite Single Lap Bonded Joints Using Digital Image Correlation, *Int. J. Adhes. Adhes.*, 2013, **40**, p 215–223.
42. M. Kashfuddoja and M. Ramji, Whole-Field Strain Analysis and Damage Assessment of Adhesively Bonded Patch Repair of CFRP Laminates Using 3D-DIC and FEA, *Compos. B Eng.*, 2013, **53**, p 46–61.
43. Shrestha, S. S. (2015). Evaluation of Composite Adhesive Bonds Using Digital Image Correlation. Southern Illinois University at Carbondale
44. F. Stuparu, D.M. Constantinescu, D.A. Apostol, and M. Sandu, A Combined Cohesive Element—XFEM Approach for Analyzing Crack Propagation in Bonded Joints, *J. Adhes.*, 2016, **92**(7–9), p 535–552.
45. J. Blaber, B. Adair, and A. Antoniou, Ncorr: Open-Source 2D Digital Image Correlation Matlab Software, *Exp. Mech.*, 2015, **55**(6), p 1105–1122.
46. R.D.S.G. Campilho, M.D. Banea, J.A.B.P. Neto, and L.F.M. da Silva, Modelling of Single-Lap Joints Using Cohesive Zone Models: Effect of the Cohesive Parameters on the Output of the Simulations, *J. Adhes.*, 2012, **88**, p 513–33. <https://doi.org/10.1080/00218464.2012.660834>
47. M. Reis, M. Adorna, O. Jiroušek, and A. Jung, Improving DIC Accuracy in Experimental Setups, *Adv. Eng. Mater.*, 2019, **21**(7), p 1900092.
48. D.C. Moreira and L.C. Nunes, Experimental Analysis of Bonded Single Lap Joint with Flexible Adhesive, *Appl. Adhes. Sci.*, 2014, **2**, p 1–8.
49. J. Weiland, M.Z. Sadeghi, J.V. Thomalla, A. Schiebahn, K.U. Schroeder, and U. Reisgen, Analysis of Back-Face Strain Measurement for Adhesively Bonded Single Lap Joints Using Strain Gauge, Digital Image Correlation and Finite Element Method, *Int. J. Adhes. Adhes.*, 2020, **97**, 102491
50. ASTM D3528-96: Standard Method for Strength Properties of Double Lap Shear Adhesive Joints by Tension Loading (2002)

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